UNIVERSIDADE DA CORUÑA

Rnasa Imedir

Unit 1.6

Instance-based Learning

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Machine Learning I

fic facultade de informática da coruña

The present material is suject to the licence

What's to be covered

- Instance-based Learning paradigm
- *K*-nearearest Neighbors (KNN)
  - Its most common variants
  - Problems
    - Working with uncertainty
    - Working with large datasets
  - Hybrid local models
  - Case-based reasoning (CBR)

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Instance-based learning

- The idea is to create a 'lazy' classifier
- To classify a new instance, we use those that are most similar from those that are already known
- Phases:
  - Training: It is not trained. Instead, the entire training set is stored
    - No computation is done
  - Generalization: given a new data point, a set of similar data is extracted from memory, and used to classify the new data
    - This is where all the computation is done

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Instance-based learning

- How do we define the concept of similarity?
- Classification cost?
  - All computation is done in prediction time!
    - Lazy Learning
- Advantage:
  - For each new instance we can get a different classifier
  - The description of the instances can be as complex as we need
- Disadvantages:
  - The cost of classification can be high
  - Irrelevant variables can affect the measure of similarity

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Instance-based learning

- *k*-NN (*k*-Nearest neighbour)
  - A new case is going to be classified in the most frequent class to which its k nearest neighbors belong
    - Also for Regression
  - Very simple and intuitive idea
  - Easy implementation
  - No explicit model

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Instance-based learning

- *k*-NN (*k*-Nearest neighbour)
  - All instances correspond to points in a space of dimension n (R^n)
    - Each instance is characterized by n values
      - Therefore, each instance is a point with n coordinates
  - The distance between two instances can be calculated by means of the Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^{n} (x_i[r] - x_j[r])^2}$$

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Instance-based learning

- *k*-NN (*k*-Nearest neighbour)
  - Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^{n} (x_i[r] - x_j[r])^2}$$

    - If the attributes have different ranges, normalize
      - Crucial step
      - What would happen if they were not normalized?
    - This calculates the distance from one instance to the rest

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Instance-based learning

- *k*-NN (*k*-Nearest neighbour)
  - Other distance measures:
    - Manhattan:

$$\sum_{i=1}^{k} [x_i - y_i]$$

    - Minkowski :

$$\left(\sum_{i=1}^{k} ([x_i - y_i])^q \right)^{1/q}$$

    - Cosine:
      - 1 – (cosine of the angle between the two points)
        - The instances are taken as vectors.

    - These (together with Euclidean) are valid only for continuous variables

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Instance-based learning

- *k*-NN (*k*-Nearest neighbour)
  - Other distance measures:
    - For categorical variables
      - Hamming distance

        - Also used when the
        - instances are binary data

Hamming Distance

$$D_H = \sum_{i=1}^{k} |x_i - y_i|$$

$$x = y \Rightarrow D = 0$$

$$x \neq y \Rightarrow D = 1$$

X	Y	Distance
Male	Male	0
Male	Female	1

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Instance-based learning

- *k*-NN (*k*-Nearest neighbour)
  - Other distance measurements:
    - Correlation
      - 1 – (correlation between instances)
        - In this case, the list of variable values that make up an instance is taken as a sequence of values

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Instance-based learning

- The parameter k identifies how many neighbours are used for the decision
- Objective Function
  - Given an instance x, and its k neighbours: $x_1, \dots, x_k$
    - Discrete case: Classification:
    - The most common value among the nearest k is assigned:
    - $f: R^n \rightarrow V,\ V = \{v_1, \dots, v_s\}$

$$\hat{f}(x) = argmax_{\ v \in V} \sum_{i=1}^{k} \delta(v, f(x_i))$$

    - where $\delta(a, b) = 1$ if $a = b$, and $\delta\ (a, b) = 0$ otherwise
      - That is, the label with which there is the greatest coincidence:

Class v such that $\sum_{i=1}^{k} \delta(v, f(x_i))$ is the highest

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Instance-based learning

- The parameter k identifies how many neighbours are used for the decision
- Objective Function
  - Given an instance x, and its k neighbours: $x_1, \ldots, x_k$
  - Continuous case: Regression:
    - The mean value of the k closest is assigned:
    - $f: R^n \rightarrow R$

$$\hat{f}(x) = \frac{\sum_{i=1}^{k} f(x_i)}{k}$$

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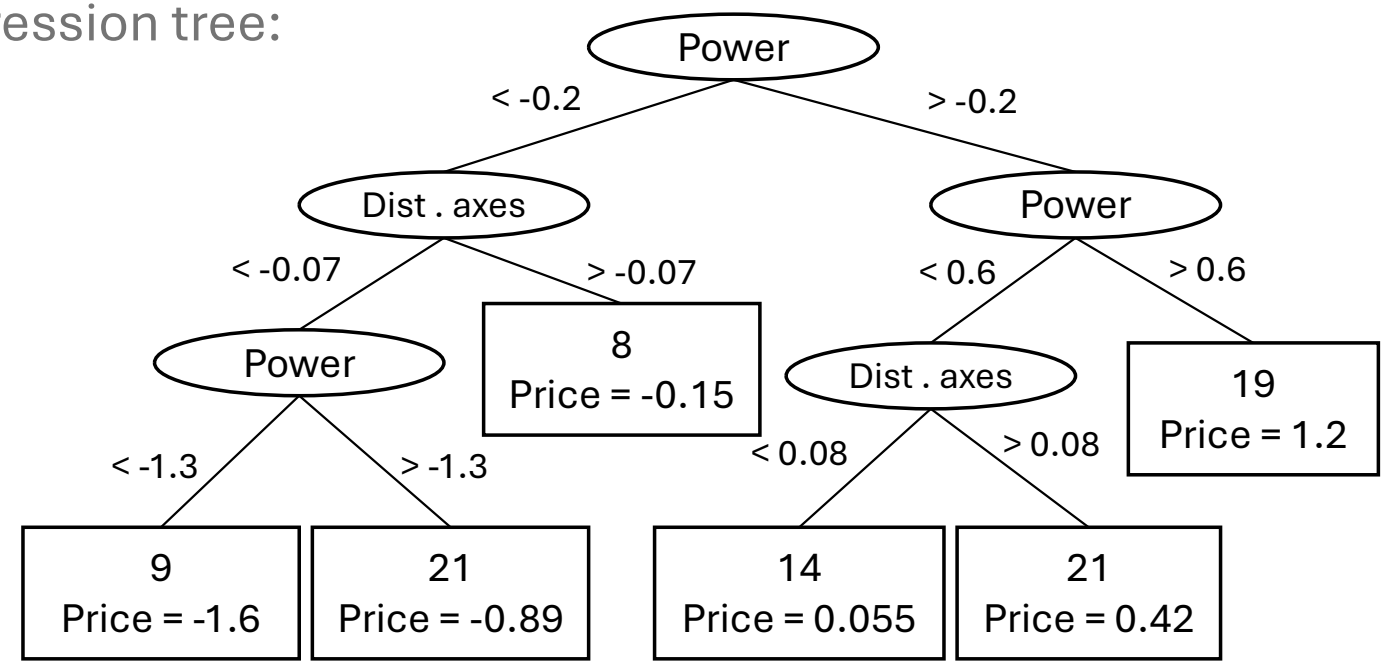
Instance-based learning

- Example: continuous case:
  - Car prices based on power and dist. axes:
  - Regression tree:



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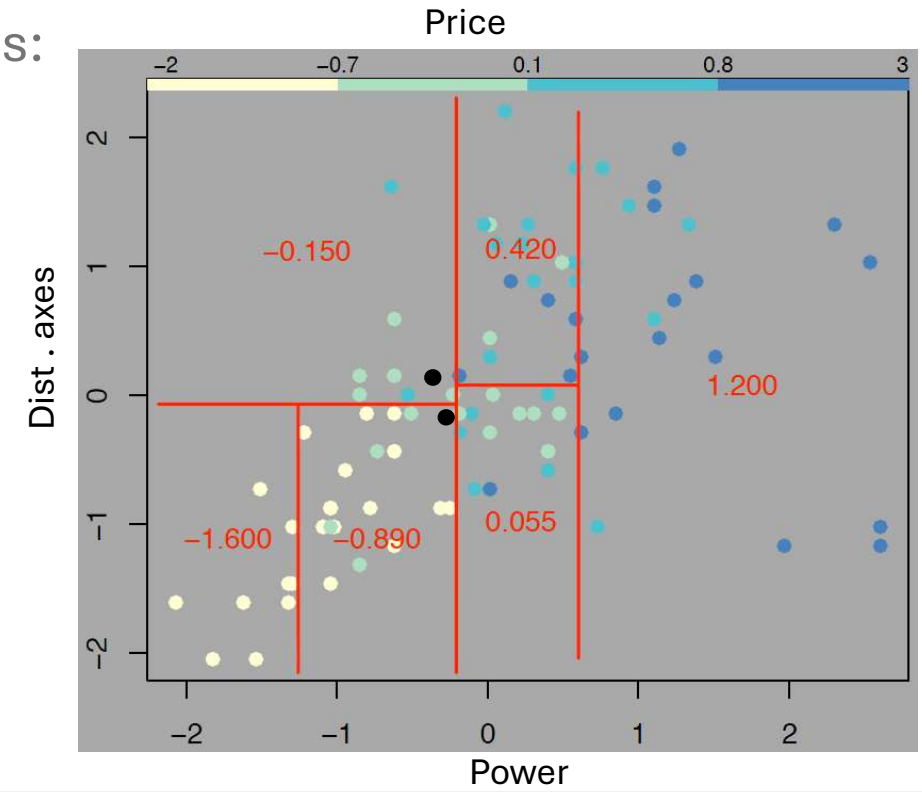
Instance-based learning

- Example: continuous case:
  - Car prices based on power and dist. axes:
  - Regression tree:

New instances:

Resulting values: -0.150 and -0.89

Very large discontinuity



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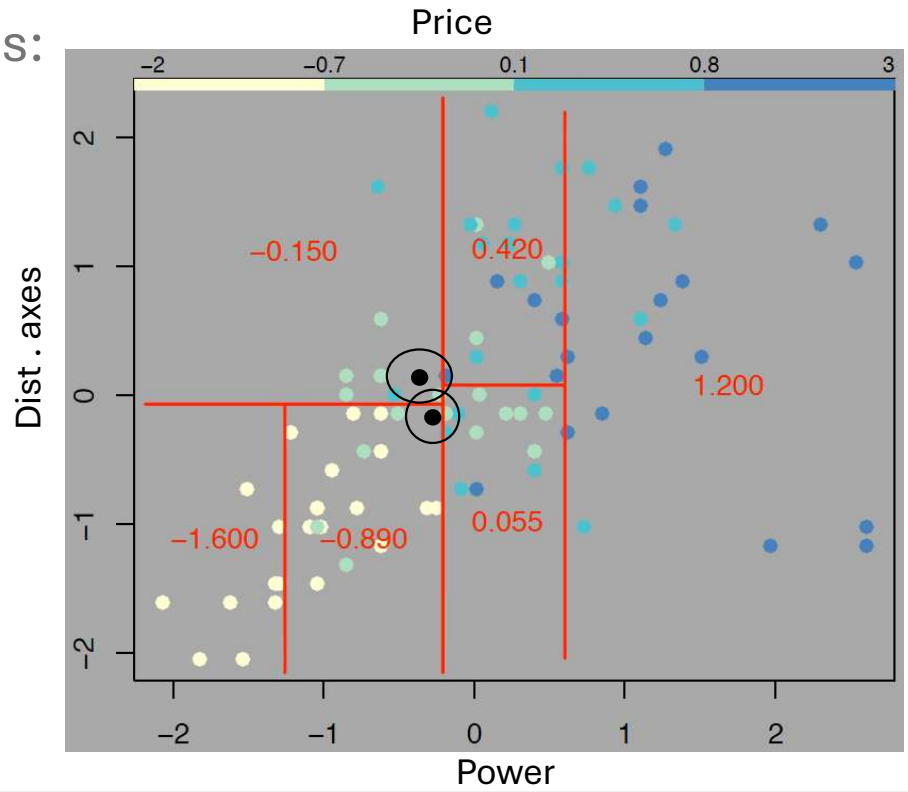
Instance-based learning

- Example: continuous case:
  - Car prices based on power and dist. axes:
  - kNN (k=3):

New instances:

Resulting values:
Average of the values of
the 3 closest instances

Value with less
discontinuity



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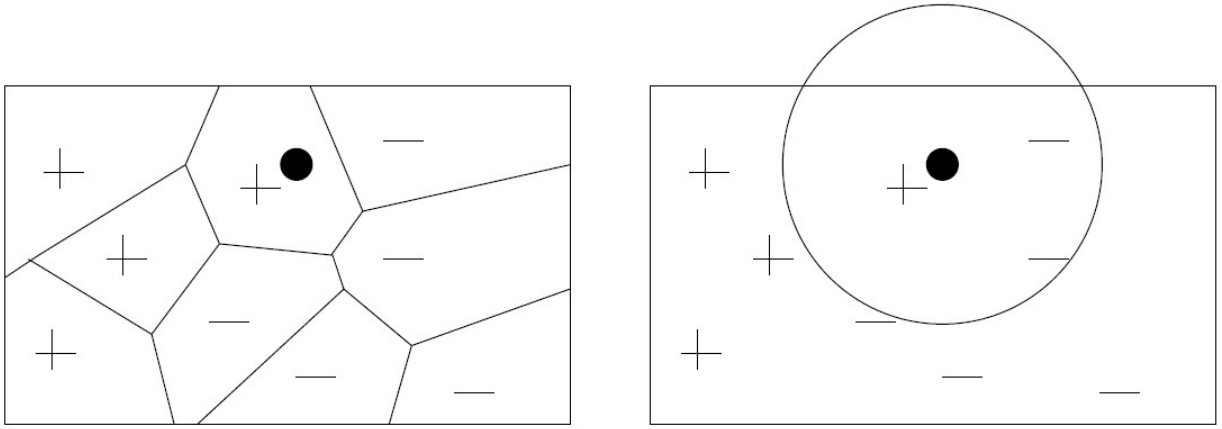
Instance-based learning

- The parameter *k* identifies how many neighbours are used for the decision



1-NN 3-NN

- Which k is chosen?
  - Trial and error process, since it depends on the problem
  - Regions formed with 1-NN are called Voronoi regions.

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Instance-based learning

- k-NN algorithm:
  - Inputs:
    - $D = \{(x_1, c_1), \dots, (x_N, c_N)\}$ dataset
    - x: new case to classify
    - k
  - k-NN(D, x, k)
    - For every object (x_i, c_i) in D,
      - Compute $d_i = d(x_i, x)$
    - Sort di in ascending order
    - Take the k cases D_x^k already classified closest to x
    - Assign to to x the most frequent class in D_x^k

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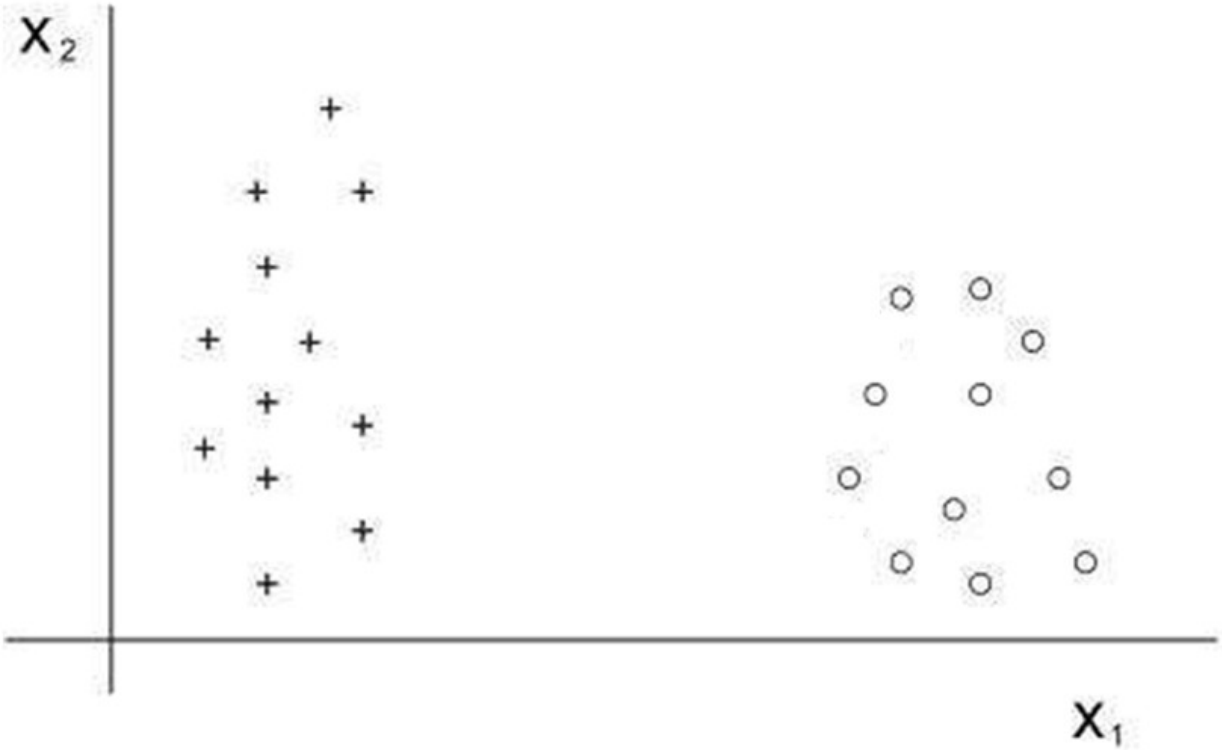
Instance-based learning

- *k*-NN: Example:



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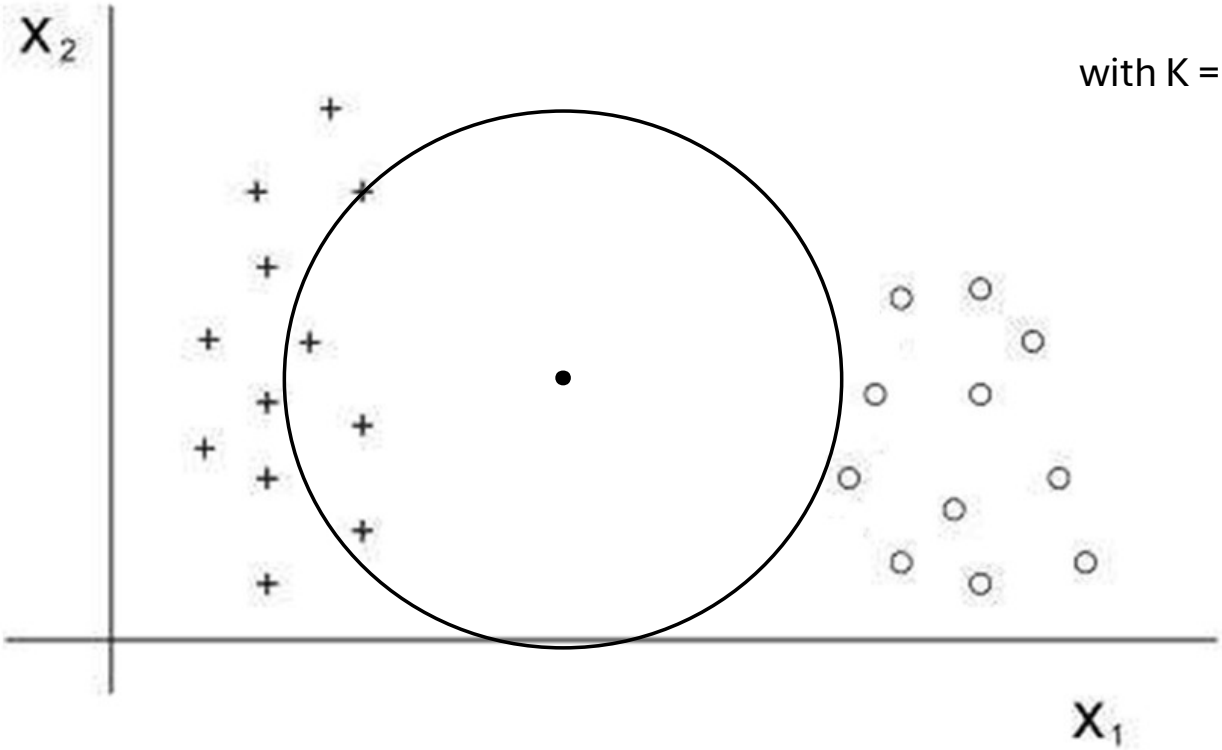
Instance-based learning

- *k*-NN: Example:



with K = 3: “+”

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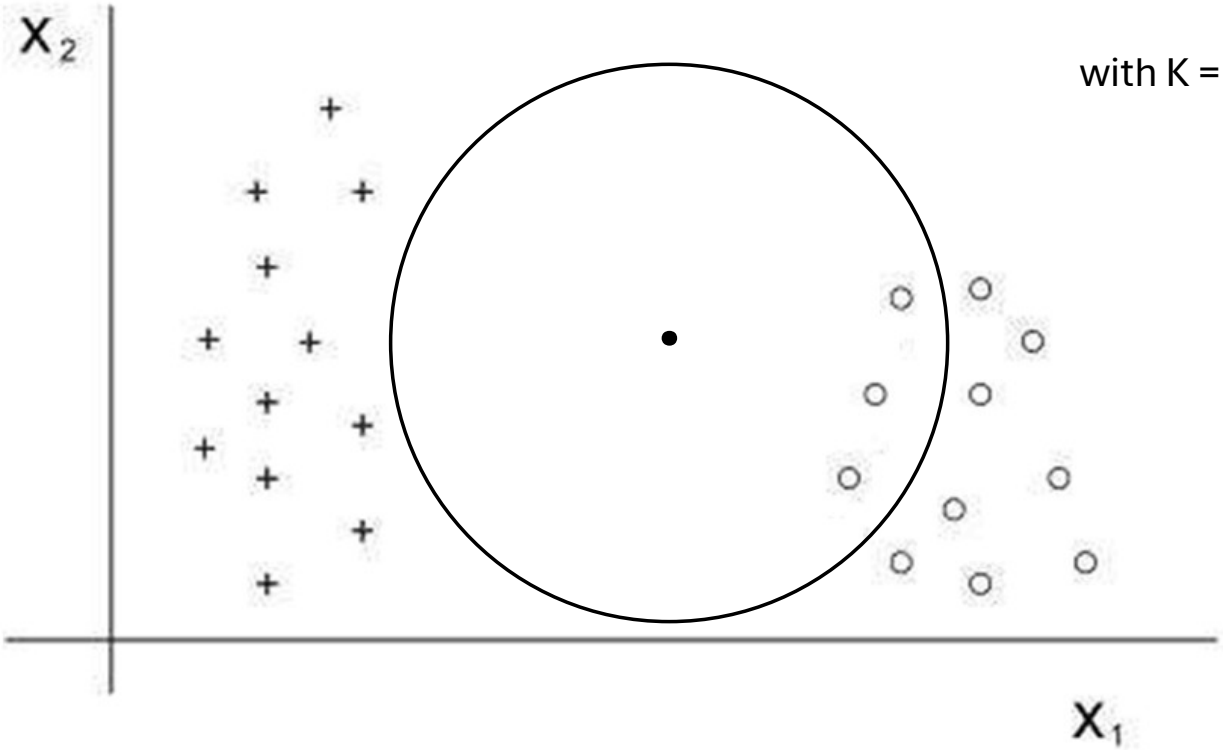
Instance-based learning

- *k*-NN: Example:



with K = 3: “o”

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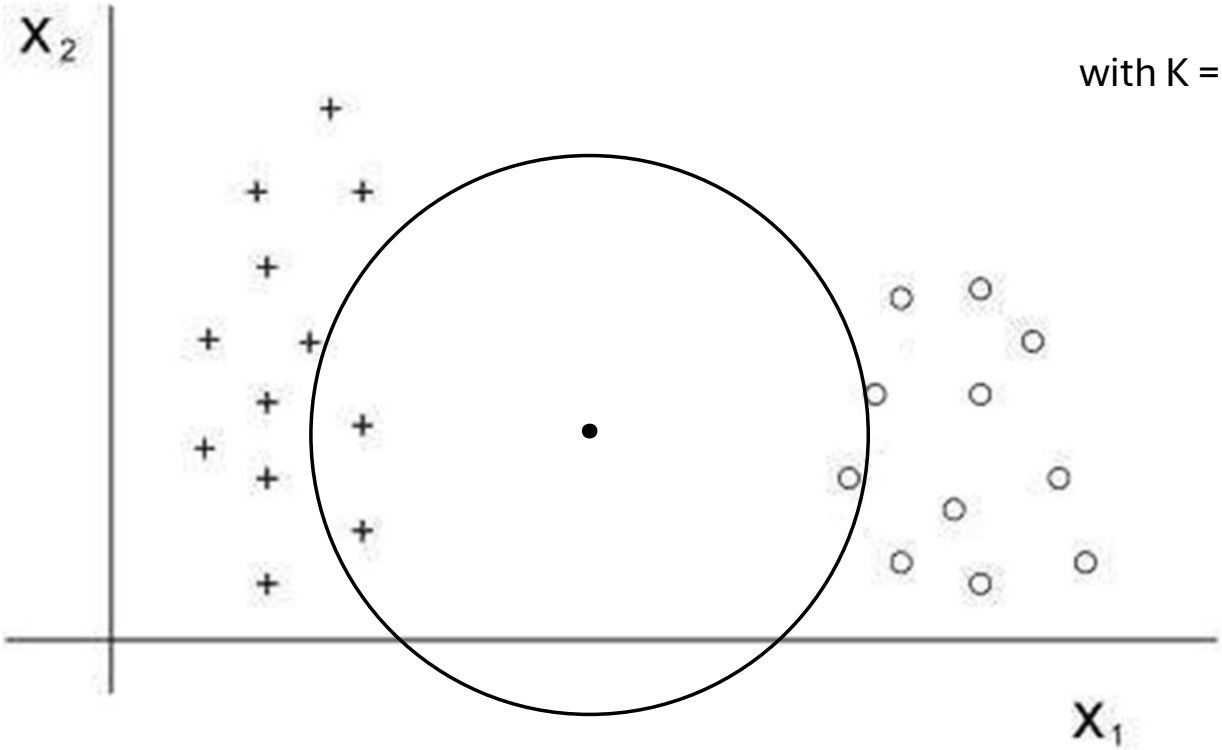
Instance-based learning

- *k*-NN: Example:



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Instance-based learning

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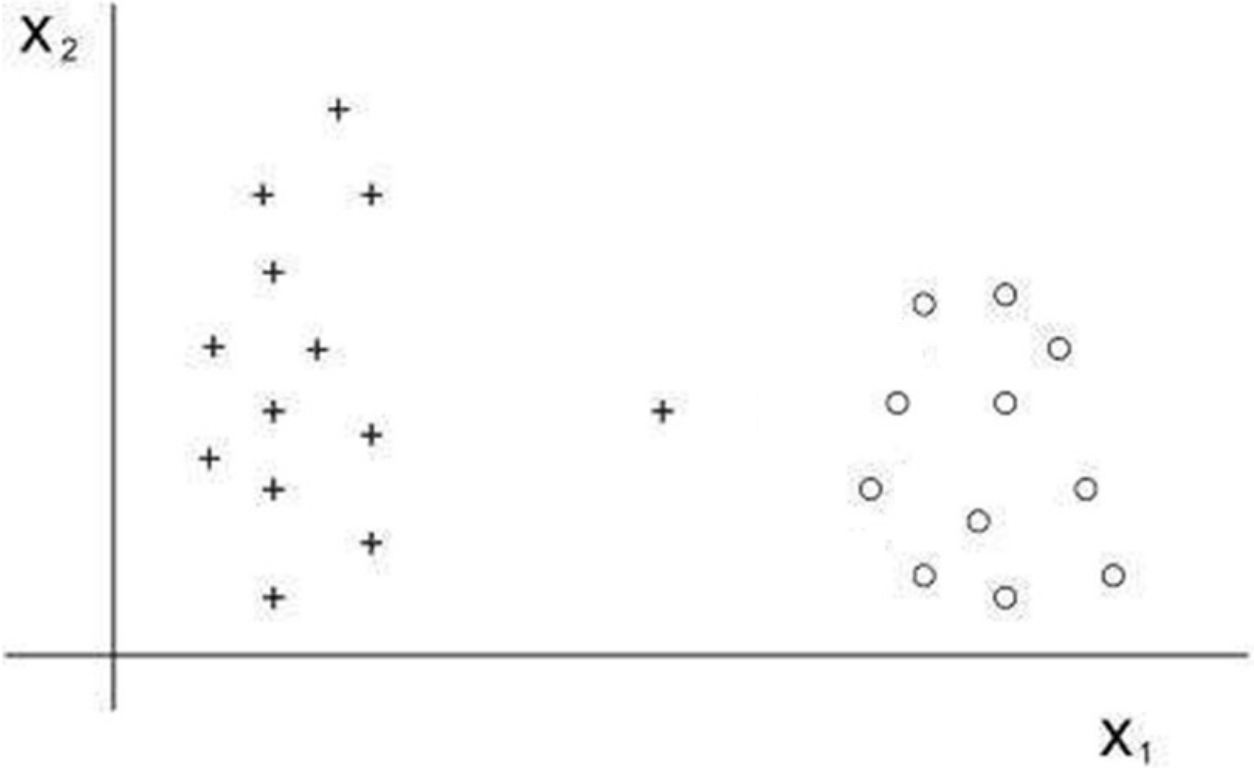

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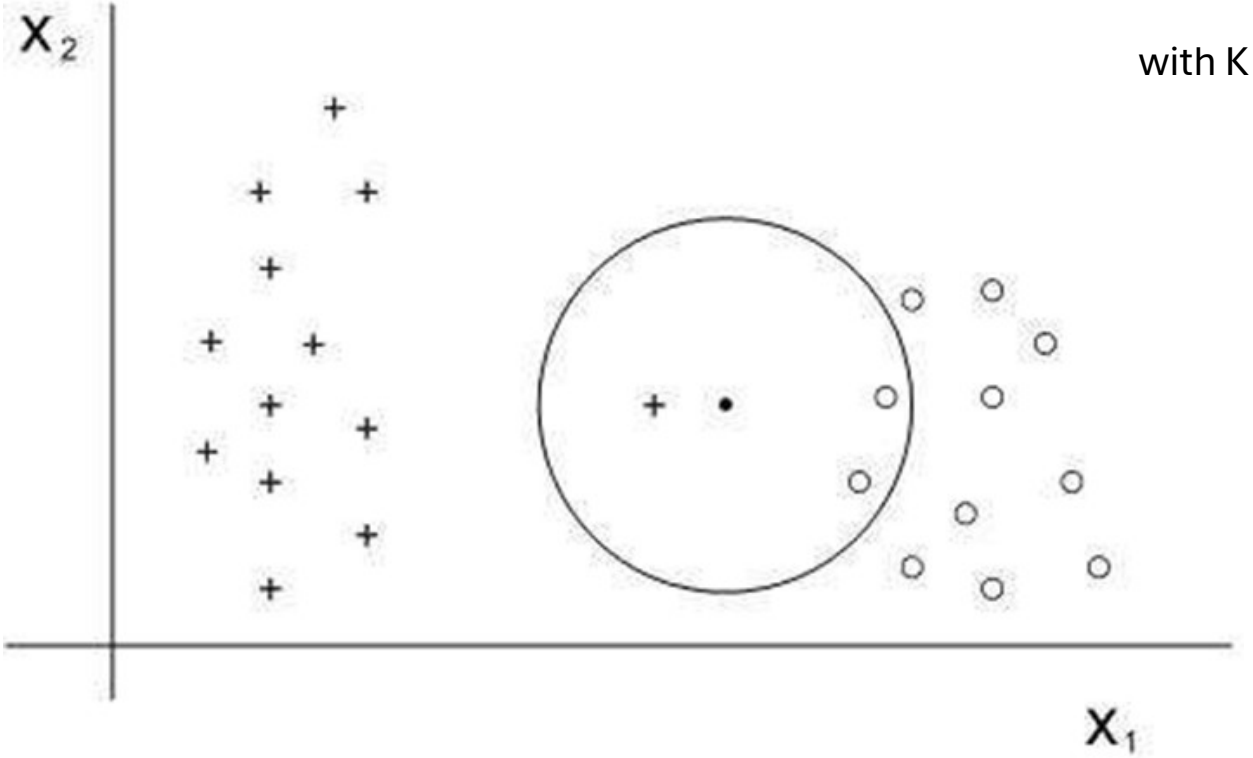
Instance-based learning

- *k*-NN: Example:



with K = 3: “o”

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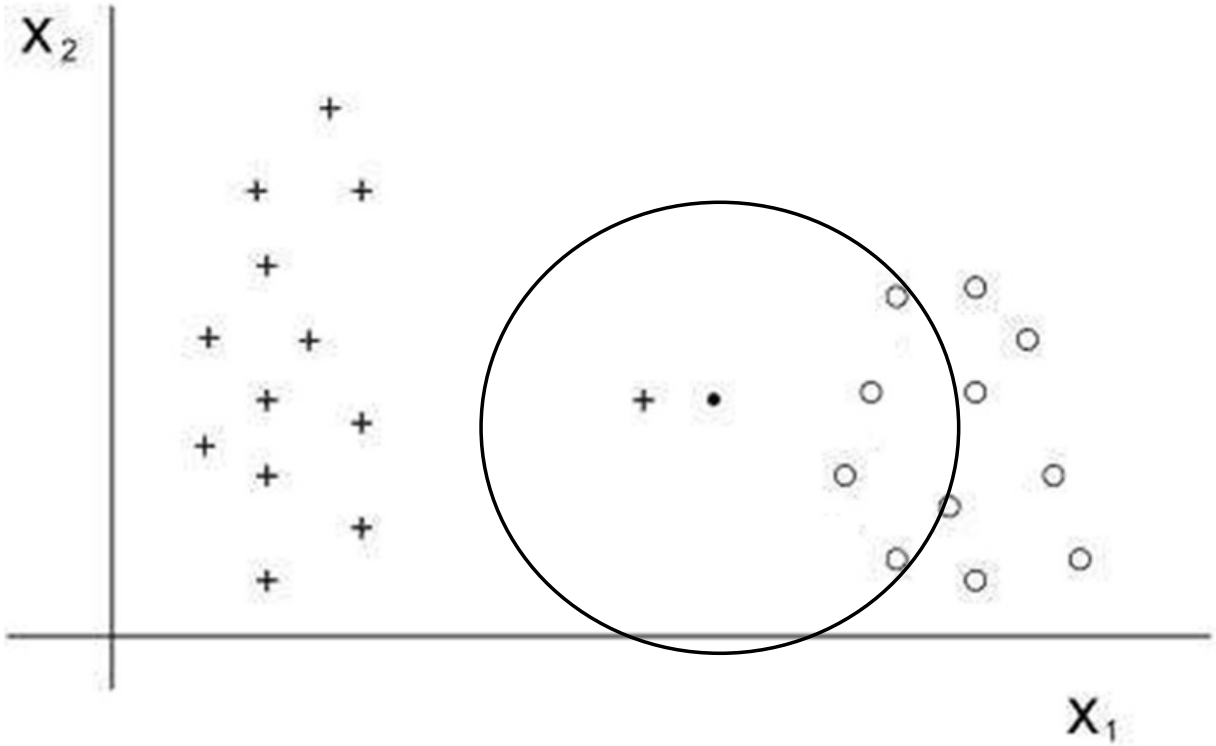
Instance-based learning

- k-NN: Example:



with K = 5: “o”

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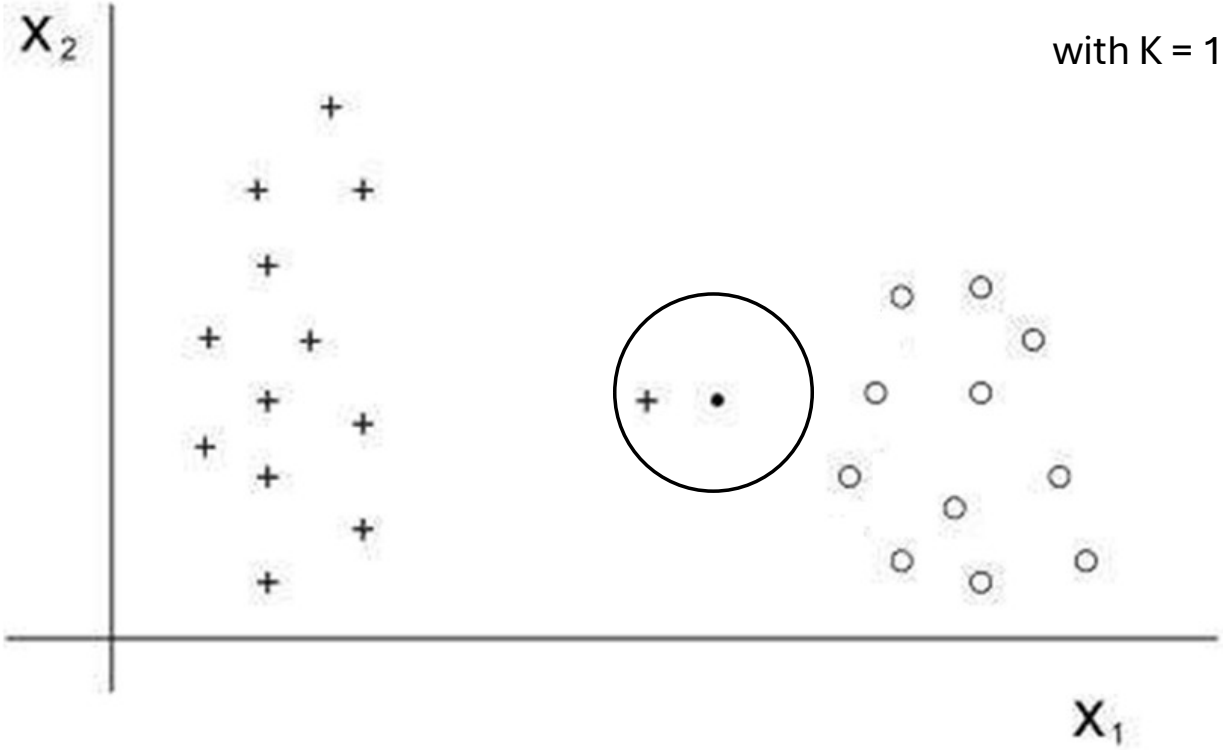
Instance-based learning

- *k*-NN: Example:



with K = 1: “+”

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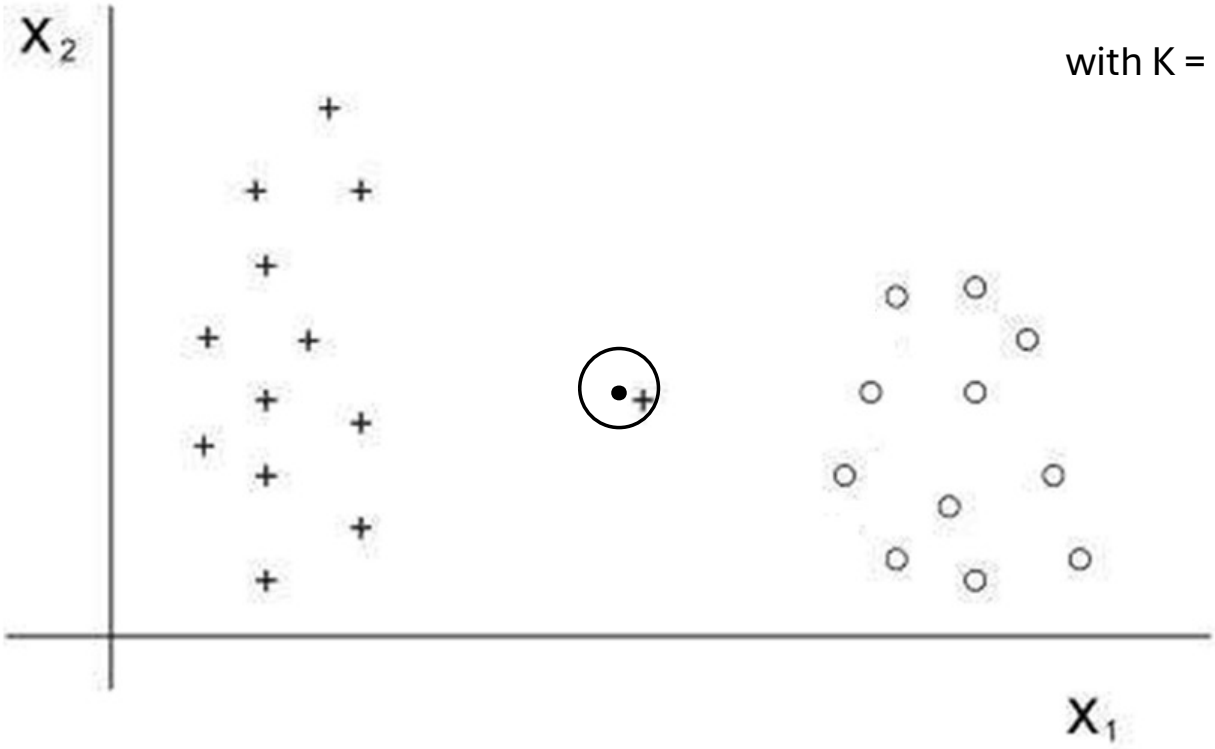
Instance-based learning

- *k*-NN: Example:



with K = 1: “+”

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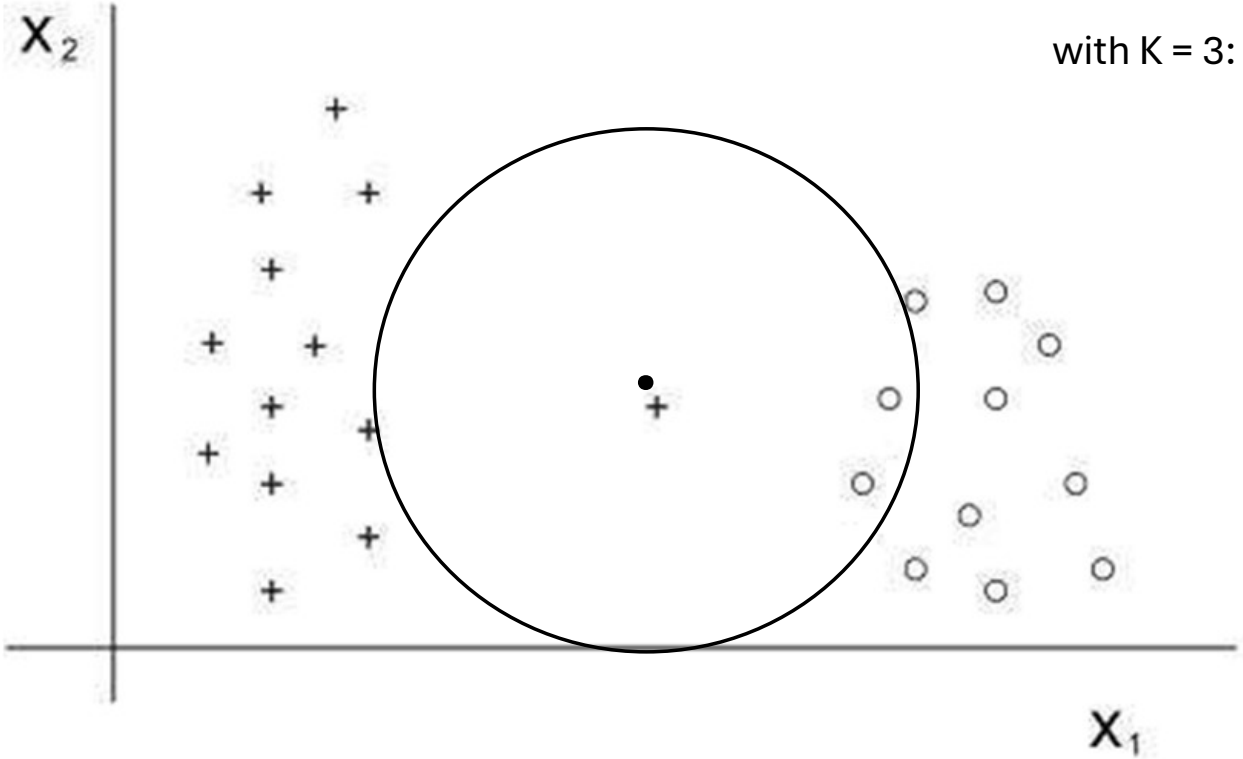
Instance-based learning

- *k*-NN: Example:



with K = 3: “o”

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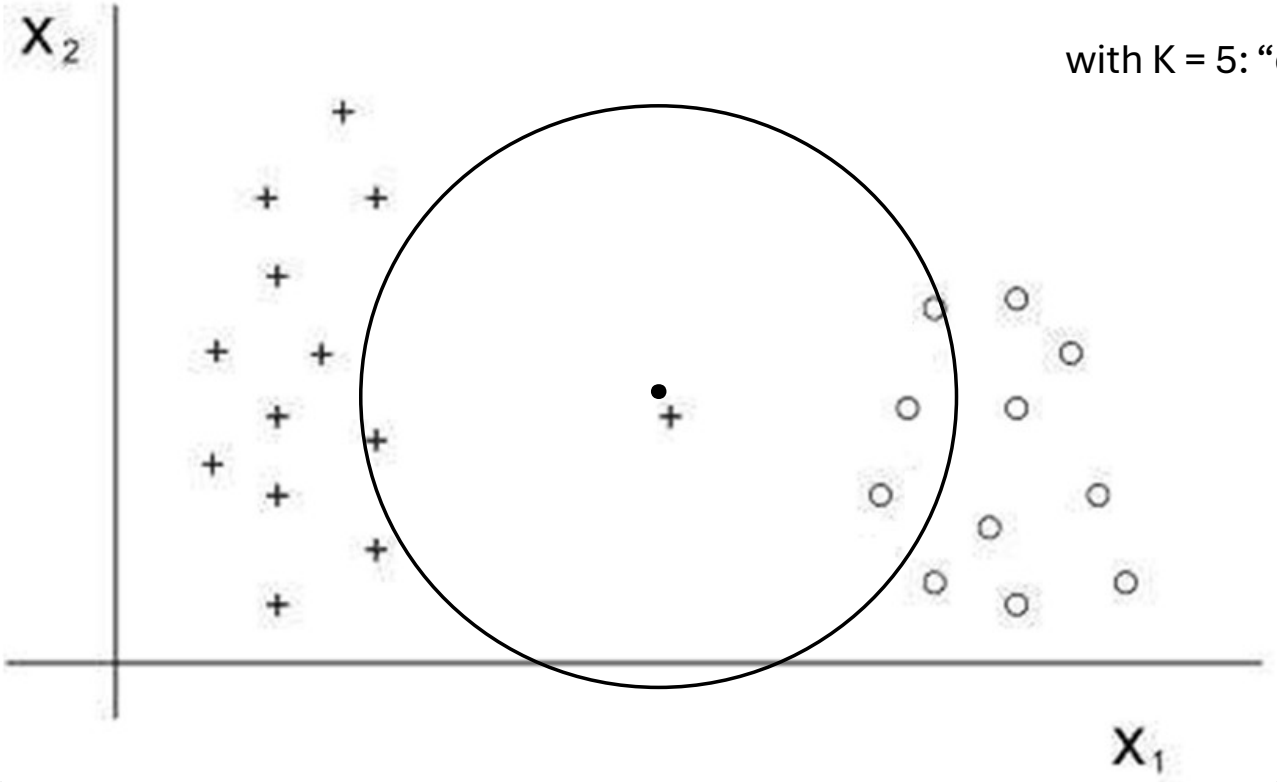
Instance-based learning

- *k*-NN: Example:



with K = 5: “o”

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Instance-based learning

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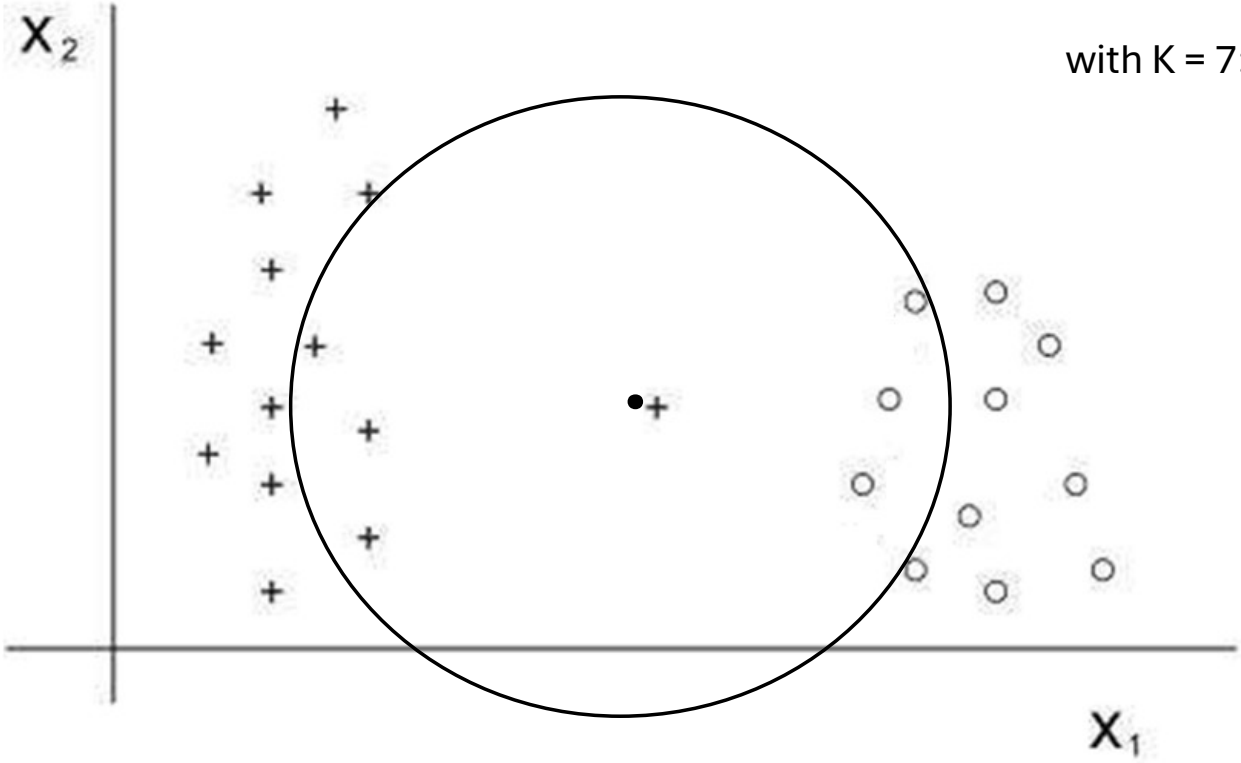

with K = 7: “+”

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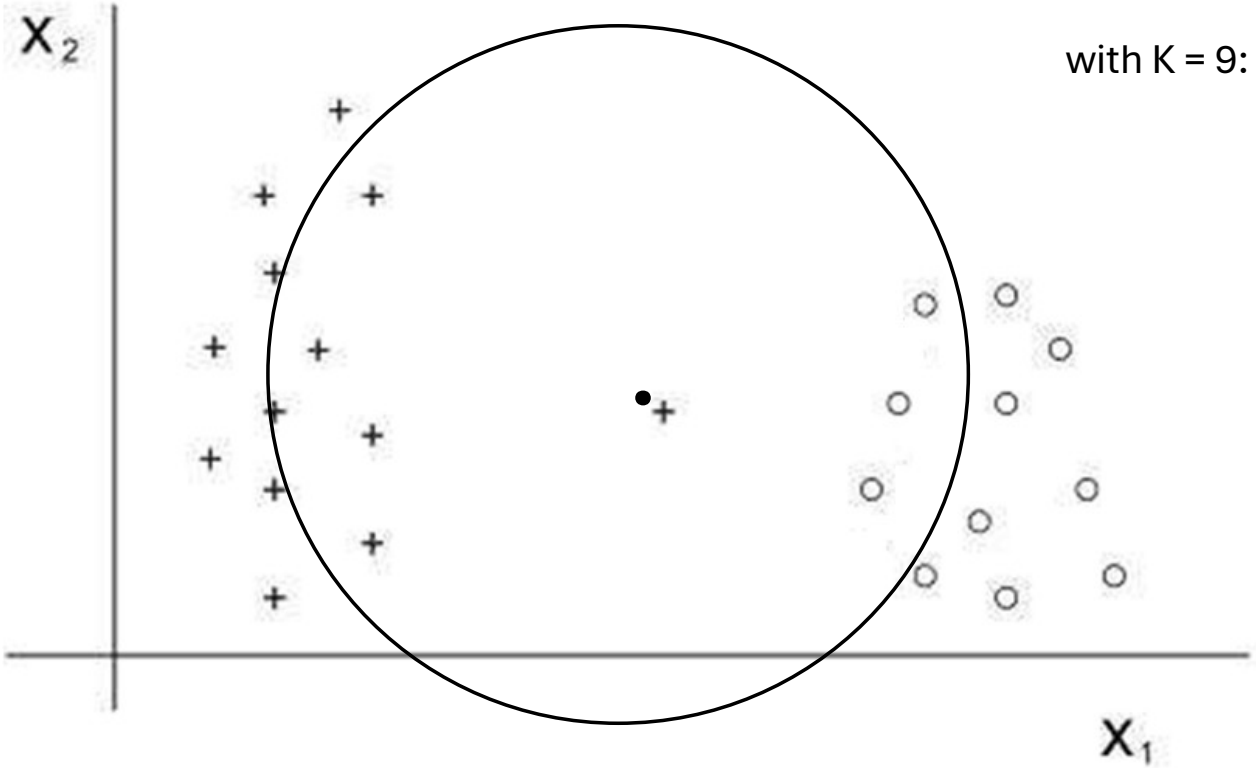
Instance-based learning

- *k*-NN: Example:



with K = 9: “+”

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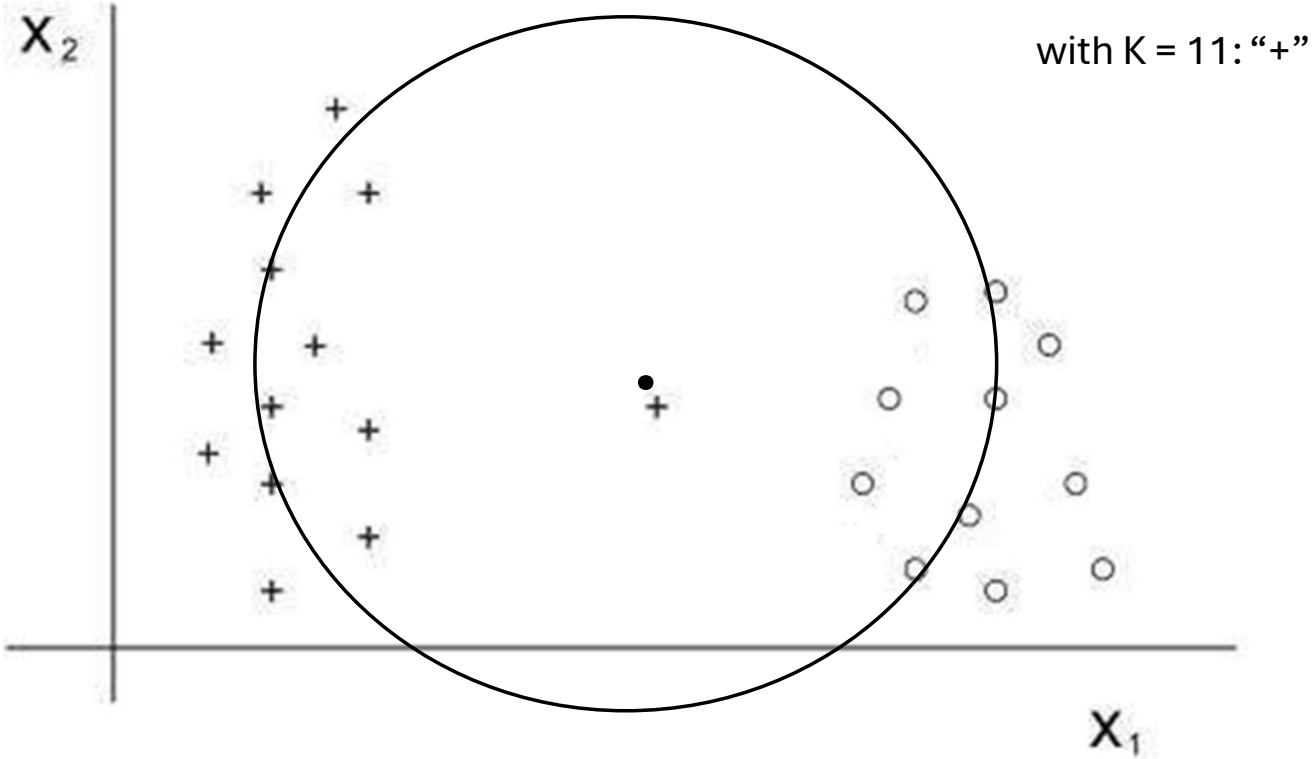
Instance-based learning

- *k*-NN: Example:



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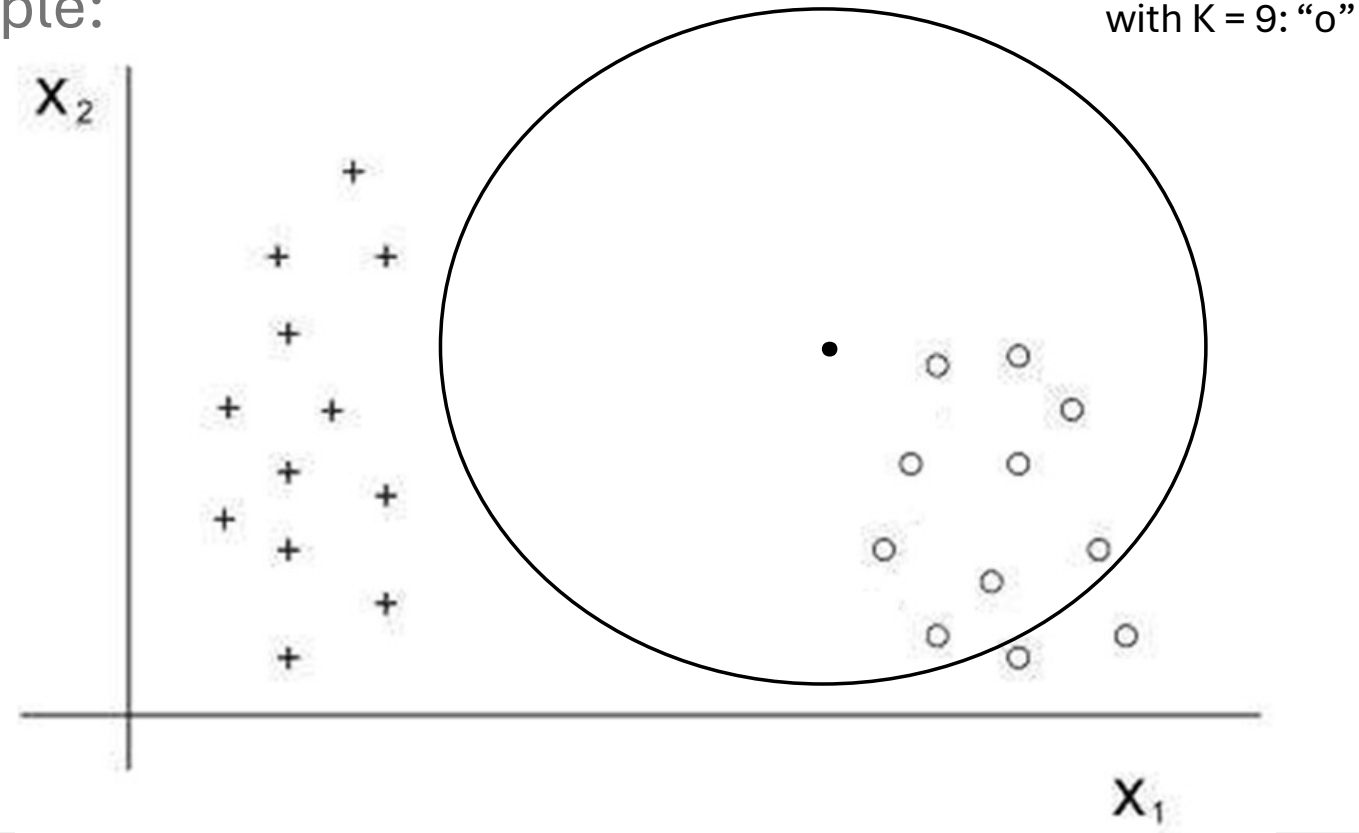
Instance-based learning

- k-NN: Example:



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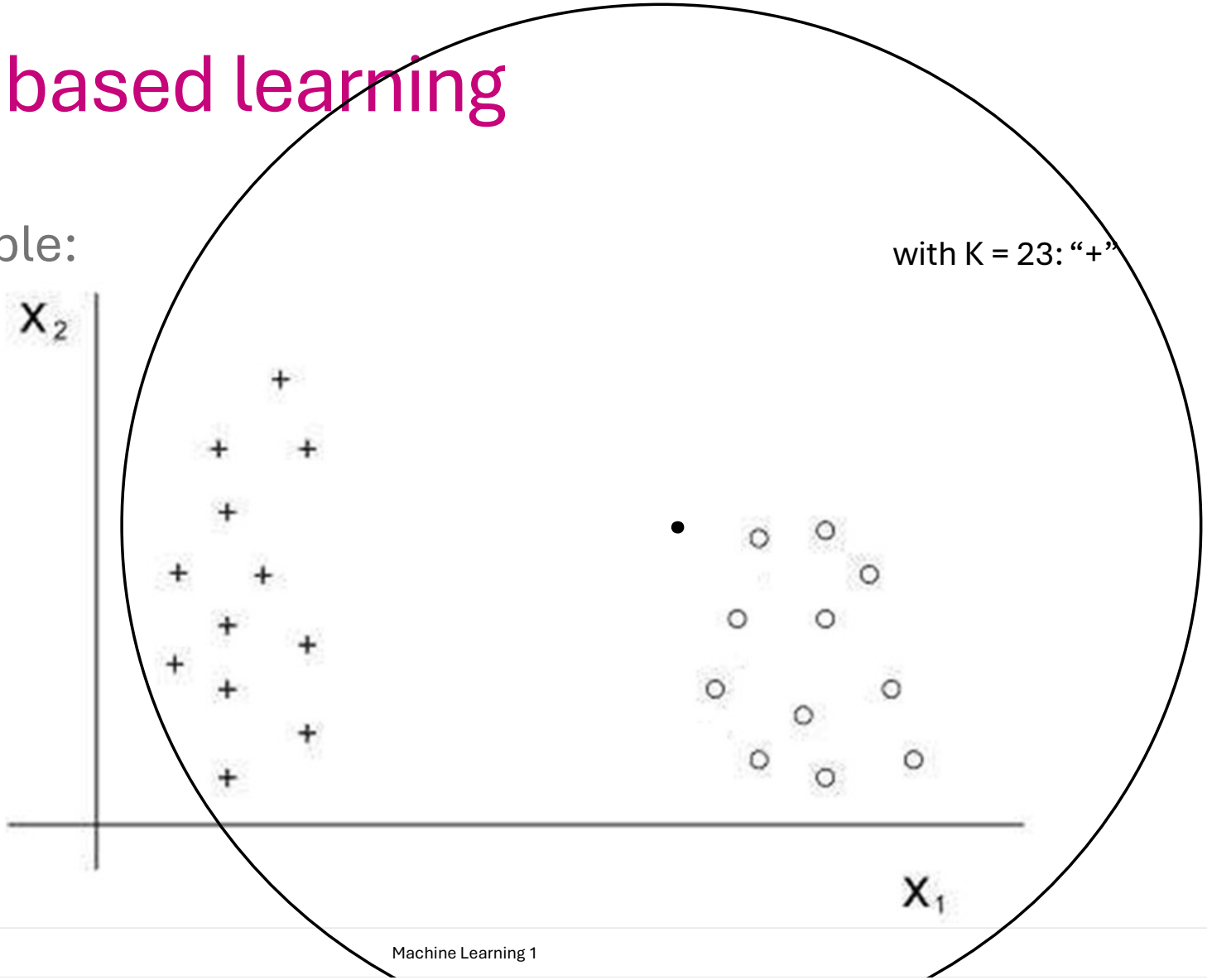
Instance-based learning

- k-NN: Example:



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Instance-based learning

- k-NN: Value of *k*:
  - If k takes a very low value, the result is very sensitive to noise
  - If it's too high, areas with many of examples can crowd out areas with fewer

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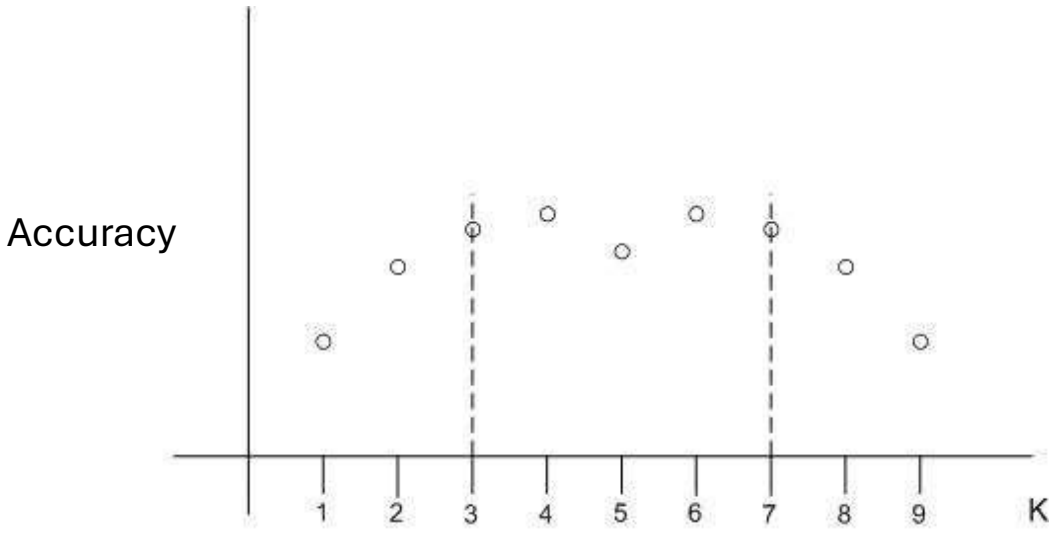
Instance-based learning

- k-NN: Value of *k*:
  - It is empirically verified that the accuracy is non-monotonic with respect to *k*
    - A good choice are usually values of *k* between 3 and 7



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Instance-based learning

- *k*-NN: Value of *k*:
  - In general, an odd k is chosen to avoid tie problems
    - The most usual values are low: 1, 3 and 5
    - For the discrete case (classification), when there are 2 classes
      - If there are more than 2 classes, a tie can be given with even and odd values
- In case there is a tie between two or more classes, it is convenient to have a heuristic rule for breaking it, for example:
  - Select the class that contains the nearest neighbor
  - Select the class with the smallest average distance
  - etc.

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Instance-based learning

- k-NN:
  - For both regression and classification, the entire space is not analyzed when issuing an output
  - Models such as SVM, ANNs, trees, etc. create a global model that is applicable in the whole space
  - In contrast, kNN analyzes a small area of the space to produce an output
    - kNN performs a local approximation
    - It only studies what the space is like in that particular area
    - The problem may be more easily solved locally than globally

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Instance-based learning

- This paradigm is somewhat atypical when compared to the rest of the classificatory paradigms:
  - In the rest of the paradigms, the classification of a new case is carried out based on two tasks:
    1. Induction of the classificatory model
    2. The subsequent deduction (or application) on the new case,
  - On the other hand, in the *k*-NN paradigm, since there is no explicit model, the two previous tasks are collapsed in what is usually called transinduction

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Instance-based learning

- Variants of *k*-NN:
  - *k*-NN with rejection
  - *k*-NN with average distance
  - *k*-NN with neighbour weighting
  - *k*-NN with minimum distance
  - *k*-NN with weighting of variables

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Instance-based learning

- Variants of *k*-NN:
  - ***k*-NN with rejection**
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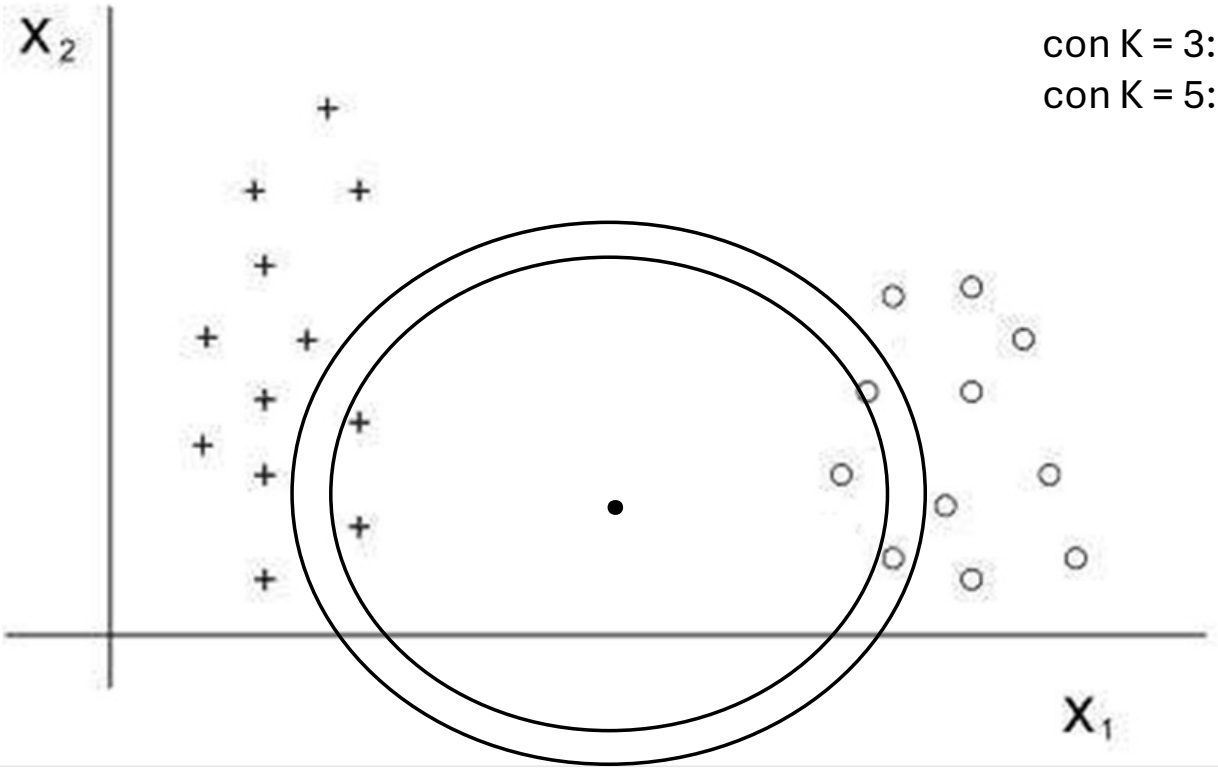
Instance-based learning

- Variants: k-NN with rejection:



con K = 3: “+”
con K = 5: “o”

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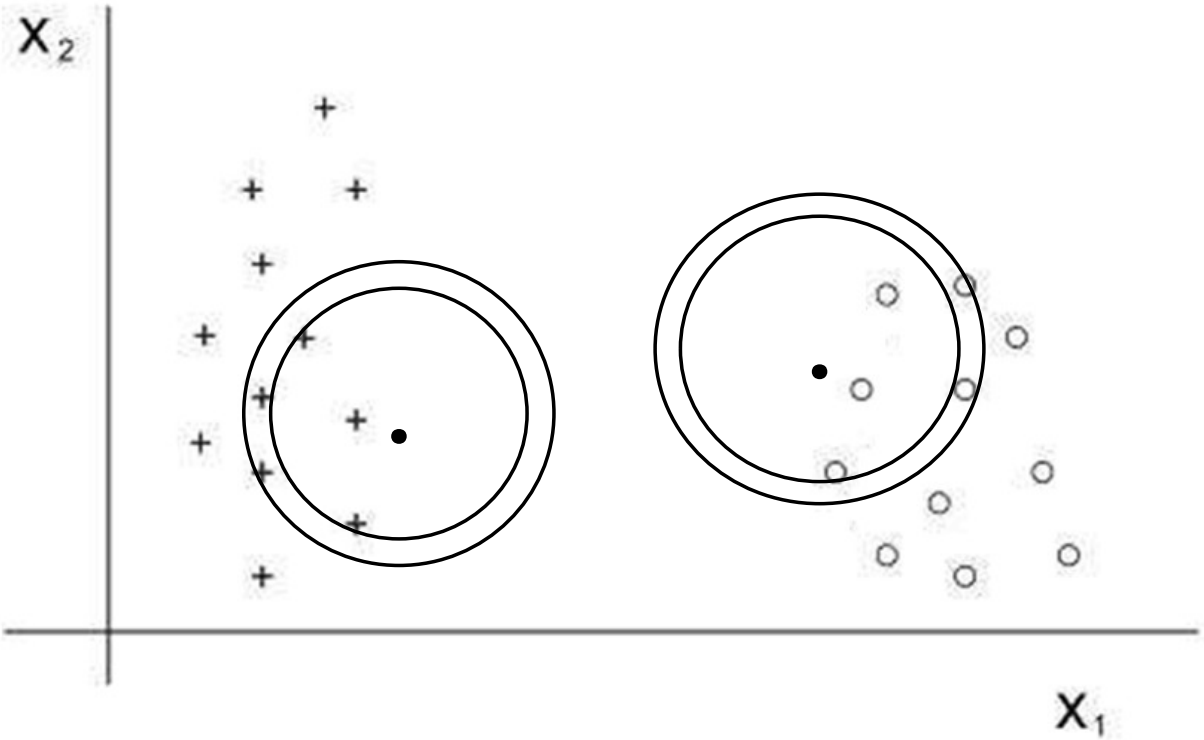
Instance-based learning

- Variants: k-NN with rejection:



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Instance-based learning

- Variants: k-NN with rejection:
  - There must be certain guarantees to classify
    - A case can remain unclassified, if there are no such guarantees that it will be assigned to the correct class
    - Examples of guarantees:
      - The number of votes obtained by the class must exceed a predetermined threshold
        - For example, if $k = 10$, with 2 classes, the threshold could be 6
      - The number of votes obtained by the most voted class must exceed the second most voted by a certain amount
        - For example, if $k = 20$, with 4 classes, a class will be assigned if the difference in votes between the most voted and the next is greater than 3
      - To assign a class, there must be an absolute majority.

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Instance-based learning

- Variants of *k*-NN:
  - *k*-NN with rejection
  - ***k*-NN with average distance**
  - *k*-NN with neighbour weighting
  - *k*-NN with minimum distance
  - *k*-NN with weighting of variables

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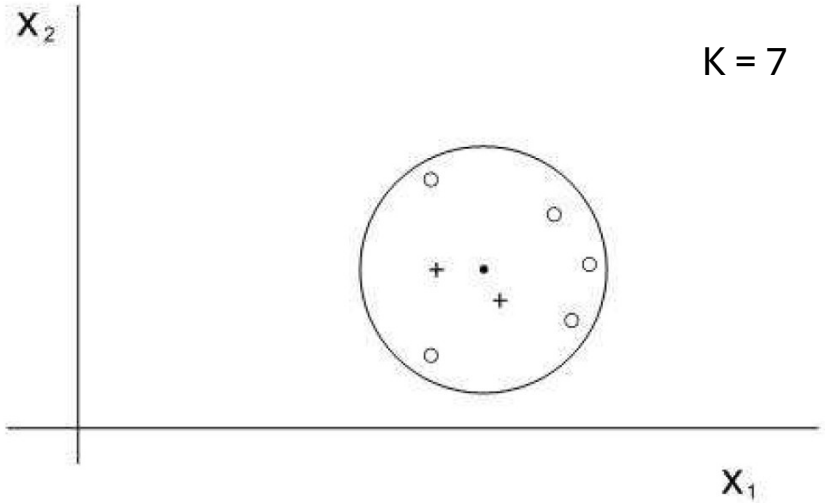
Instance-based learning

- Variants: k-NN with average distance:
  - Assign a new case to the class whose average distance is the smallest



- In this case, the new instance is classified as "+", because the average distance to the two instances of class "+" is smaller than the average distance to the 5 "o" instances

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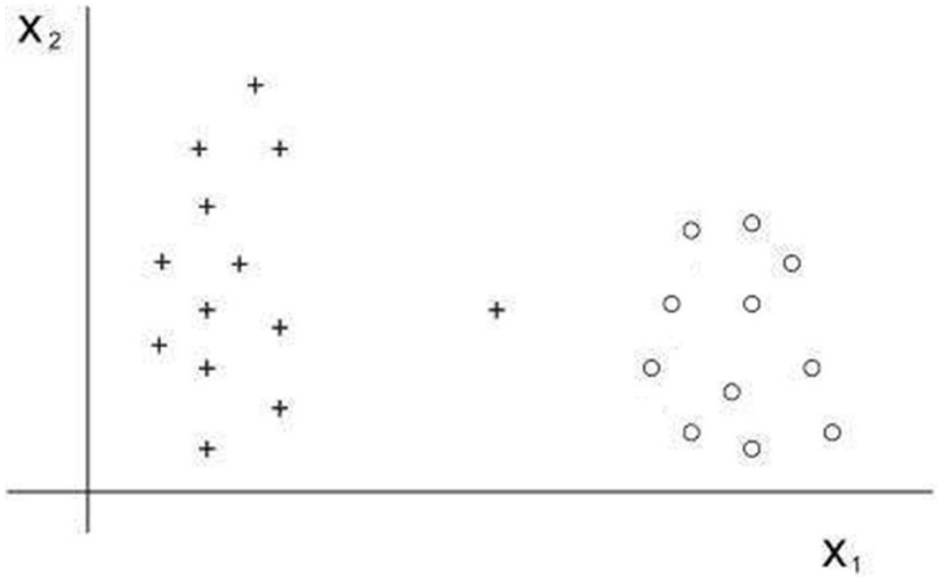
Instance-based learning

- Variants: k-NN with average distance:
  - Assign a new case to the class whose average distance is the smallest
    - Problem: too sensitive to noise: Example:



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Instance-based learning

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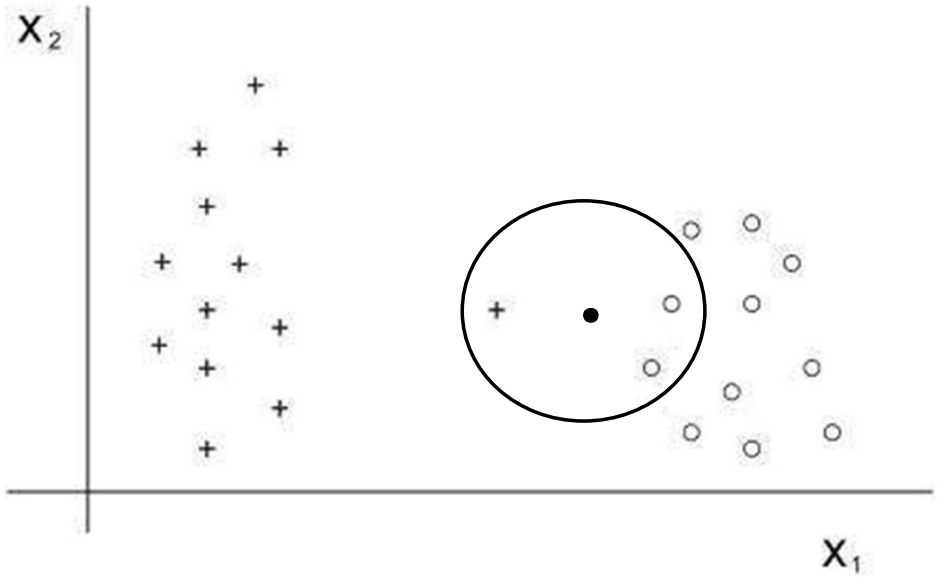

with K = 3: “+”

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Instance-based learning

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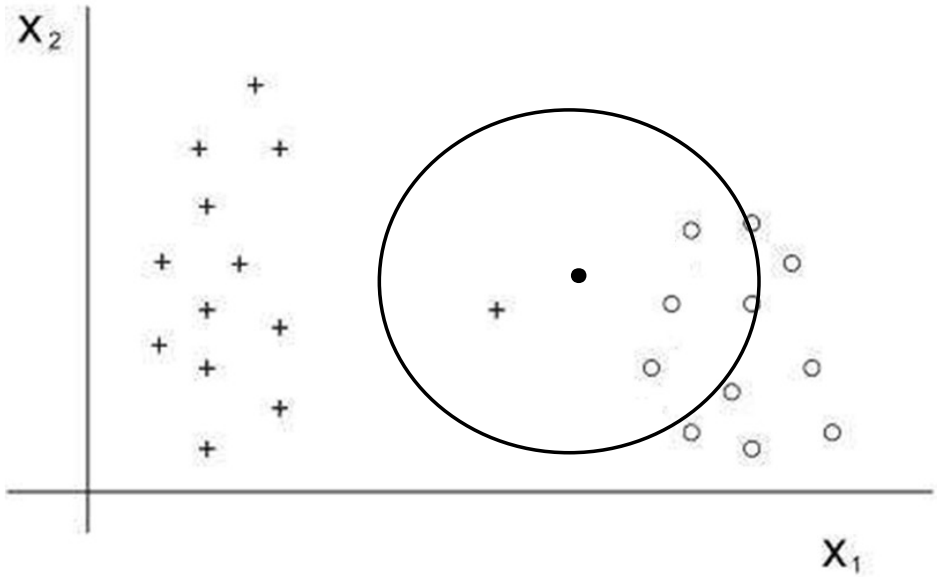

with K = 5: “+”

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Instance-based learning

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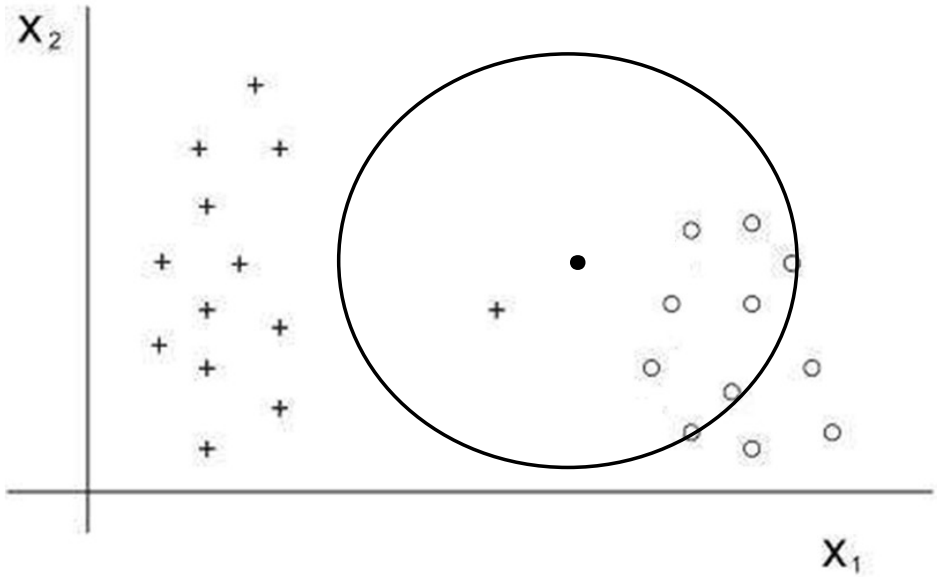

with K = 9: “+”

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Instance-based learning

- Variants of *k*-NN:
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  - *k*-NN with minimum distance
  - *k*-NN with weighting of variables

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Instance-based learning

- Variants: *k*-NN with neighbour weighting
  - Distance-Weighted *k*-NN:
  - Weigh the importance that each neighbor contributes to the value of the objective function, based on its distance
    - Example: inverse of the distance
    - Example: inverse of distance squared

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Instance-based learning

- Variants: k-NN with neighbour weighting
  - Given an instance x, and its k neighbours: $x_1, \ldots, x_k$
    - Discrete case:
      - Objective function:

$$\hat{f}(x) = argmax_{\ v \in V} \sum_{i=1}^{k} \delta(v, f(x_i))$$

      - where $w_i = 1/d(x, x_i)^2$ If any x_i exactly matches x, the unweighted version is used for those that meet that condition.
        - Or return the class of x_i

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Instance-based learning

- Variants: k-NN with neighbour weighting
  - Given an instance x, and its k neighbours: $x_1, \ldots, x_k$
    - Continuous case:
      - Objective Function:

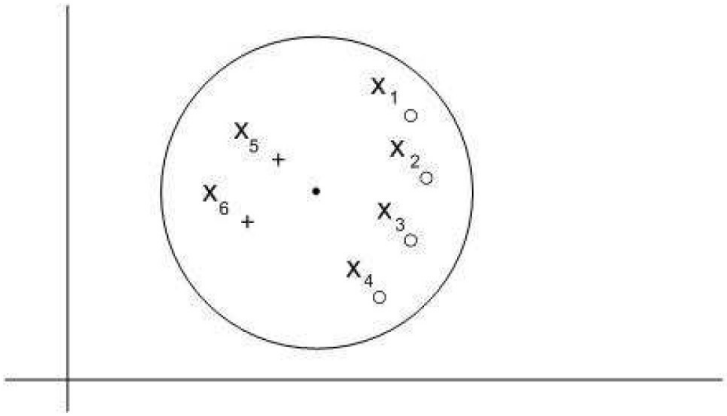
$$\hat{f}(x) = \frac{\sum_{i=1}^{k} w_i f(x_i)}{\sum_{i=1}^{k} w_i}$$

      - where $w_i = 1/d(x, x_i)^2$

The present material is suject to the licence

Instance-based learning

- Variants: *k*-NN with neighbour weighting



	$d(\mathbf{x}_i, \mathbf{x})$	w_i
$\mathbf{x}_1$	2	0,5
$\mathbf{x}_2$	2	0,5
$\mathbf{x}_3$	2	0,5
$\mathbf{x}_4$	2	0,5
$\mathbf{x}_5$	0,7	1/0,7
$\mathbf{x}_6$	0,8	1/0,8

$$w_i = \frac{1}{d(x_i, x)}$$

- In this case, the weights relative to class “o” add up to 2, and the weights relative to class “+” add up to 2.67
  - It is classified as “+” even though there are more neighbors of class “o”

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Instance-based learning

- Variants of k-NN:
  - *k*-NN with rejection
  - *k*-NN with average distance
  - *k*-NN with neighbour weighting
  - ***k*-NN with minimum distance**
  - *k*-NN with weighting of variables

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Instance-based learning

- Variants: k-NN with minimum distance:
  - Start by selecting **one case per class**
    - Generally, the closest case to the centroid of all elements of that class
    - In this way, the size of the case file to be stored is reduced from the number of examples to the number of classes
  - Given a new case to classify, this new case is assigned to the class whose representative is closest
    - It is like doing a 1-NN to the set with only one case per class

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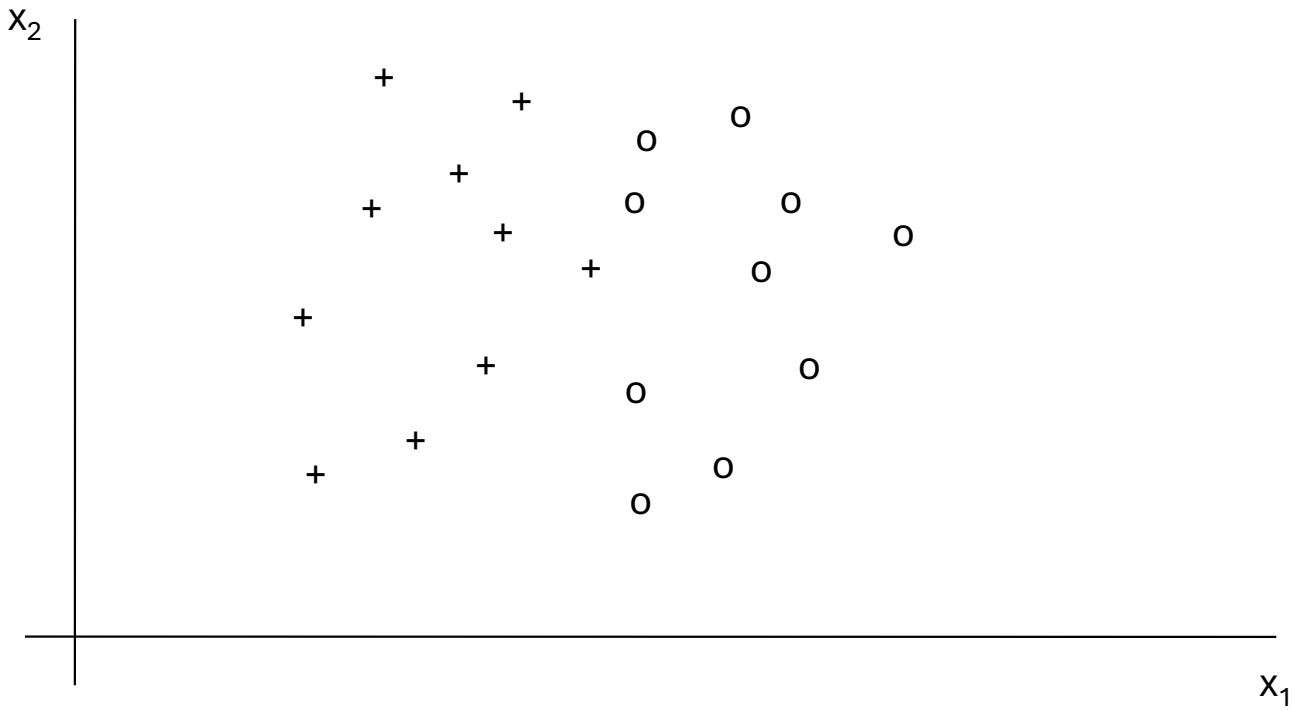
Instance-based learning

- Variants: *k*-NN with minimum distance:
  - Dataset:



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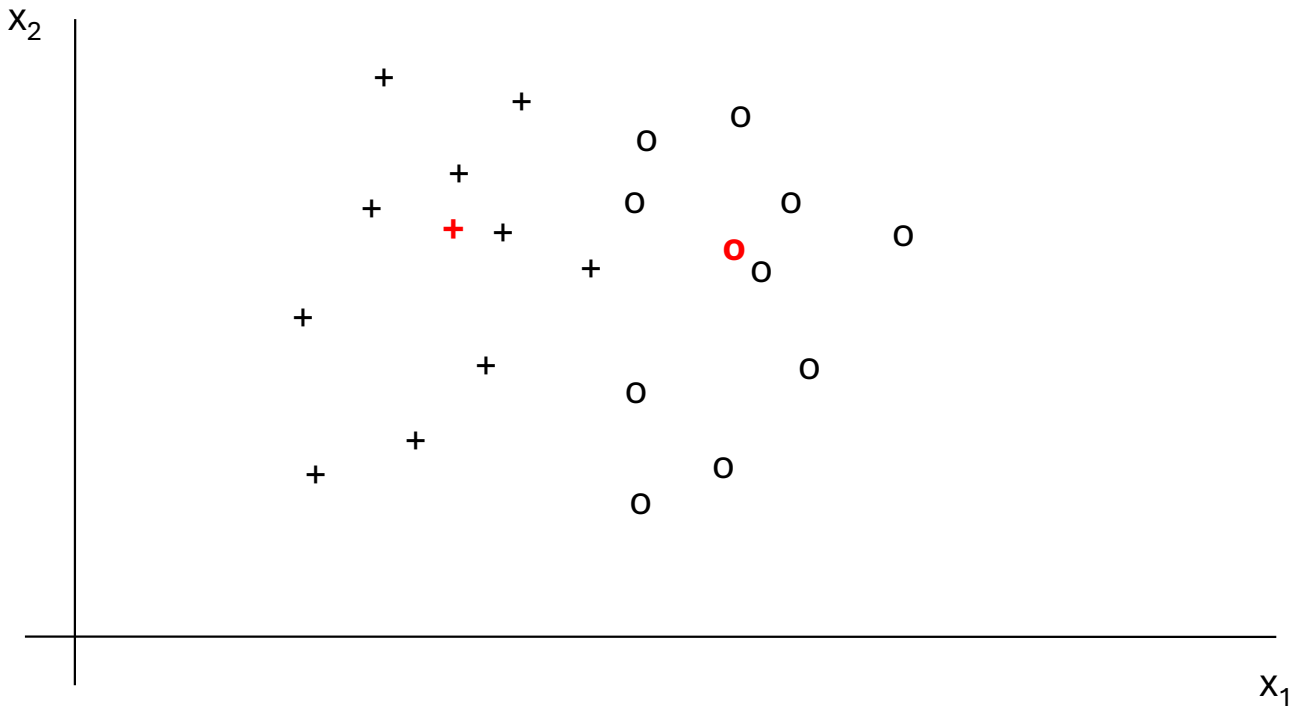
Instance-based learning

- Variants: *k*-NN with minimum distance:
  - Centroids of each class:



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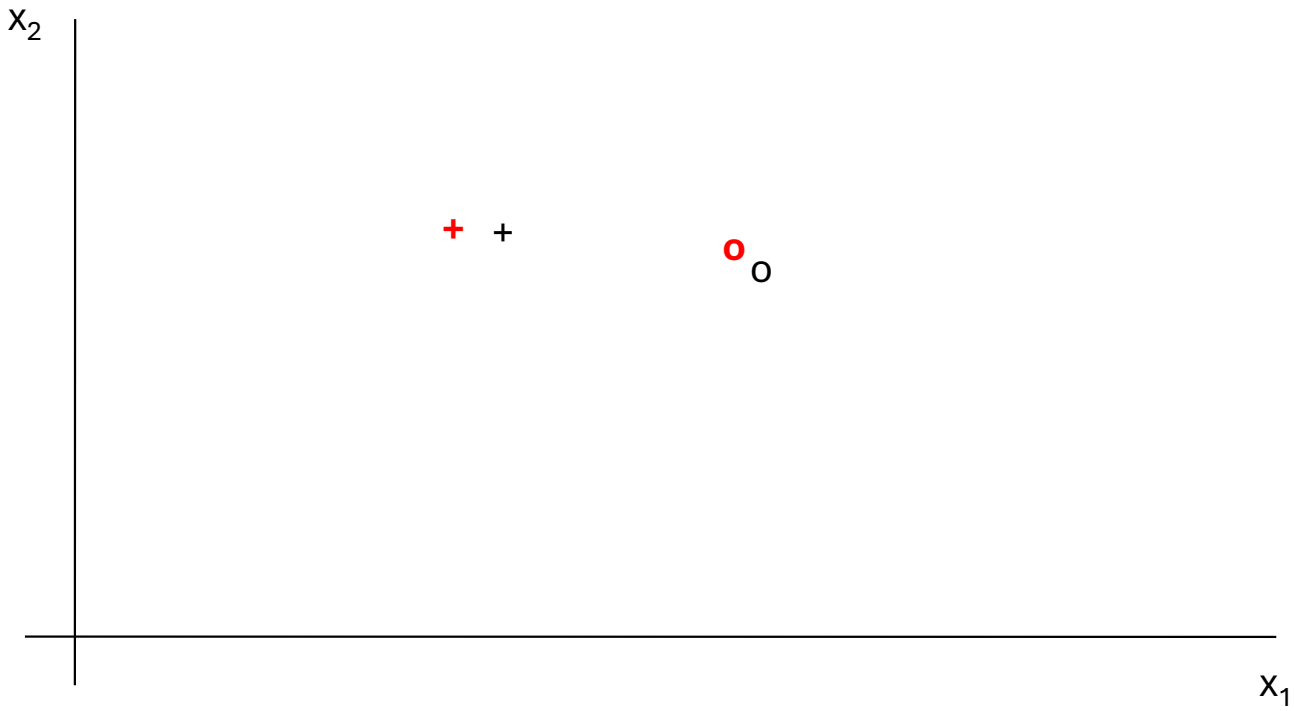
Instance-based learning

- Variants: *k*-NN with minimum distance:
  - Instances closest to the centroids:



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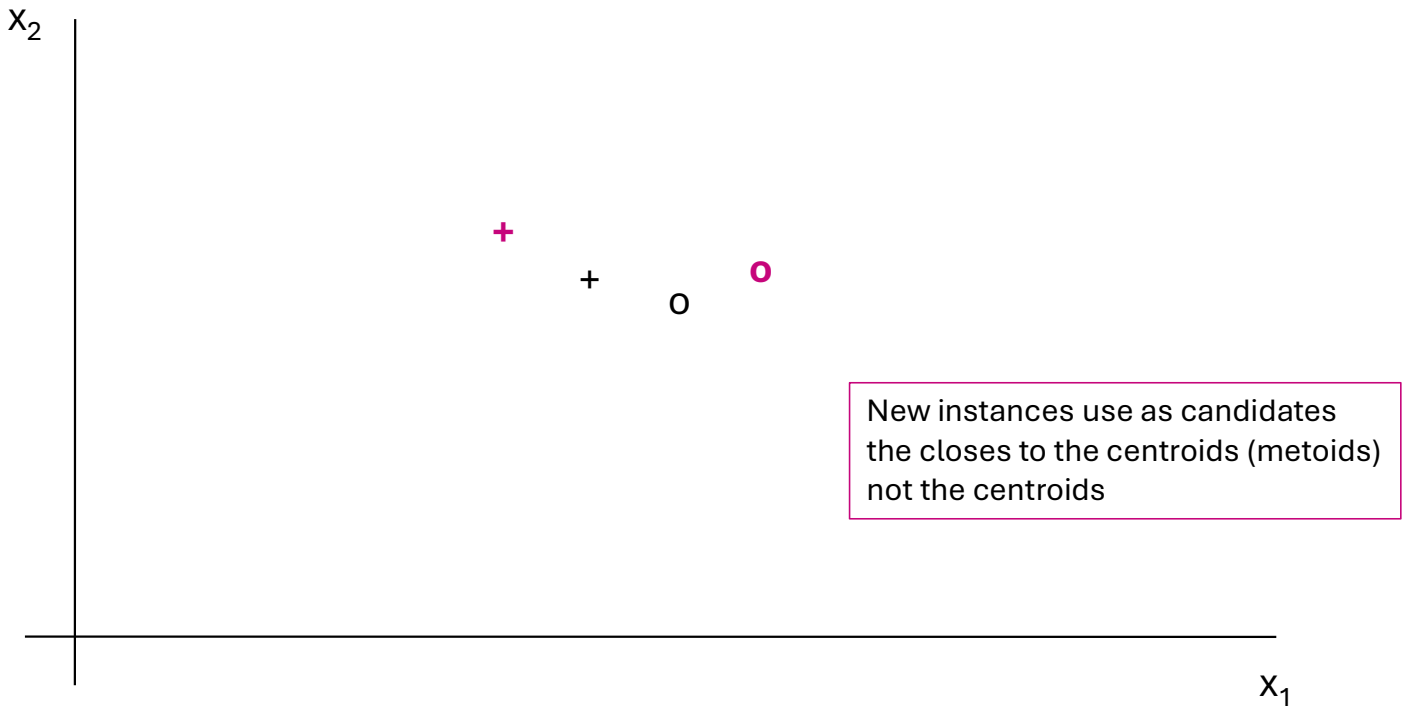
Instance-based learning

- Variants: *k*-NN with minimum distance:
  - Classification of new instances:



New instances use as candidates the closes to the centroids (metoids) not the centroids

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Instance-based learning

- Variants: *k*-NN with minimum distance:
  - Advantage:
    - Computational cost is lower than the generic *k*-NN
  - Disadvantage:
    - Its effectiveness is conditional on homogeneity within classes
      - The more homogeneous, the more effective

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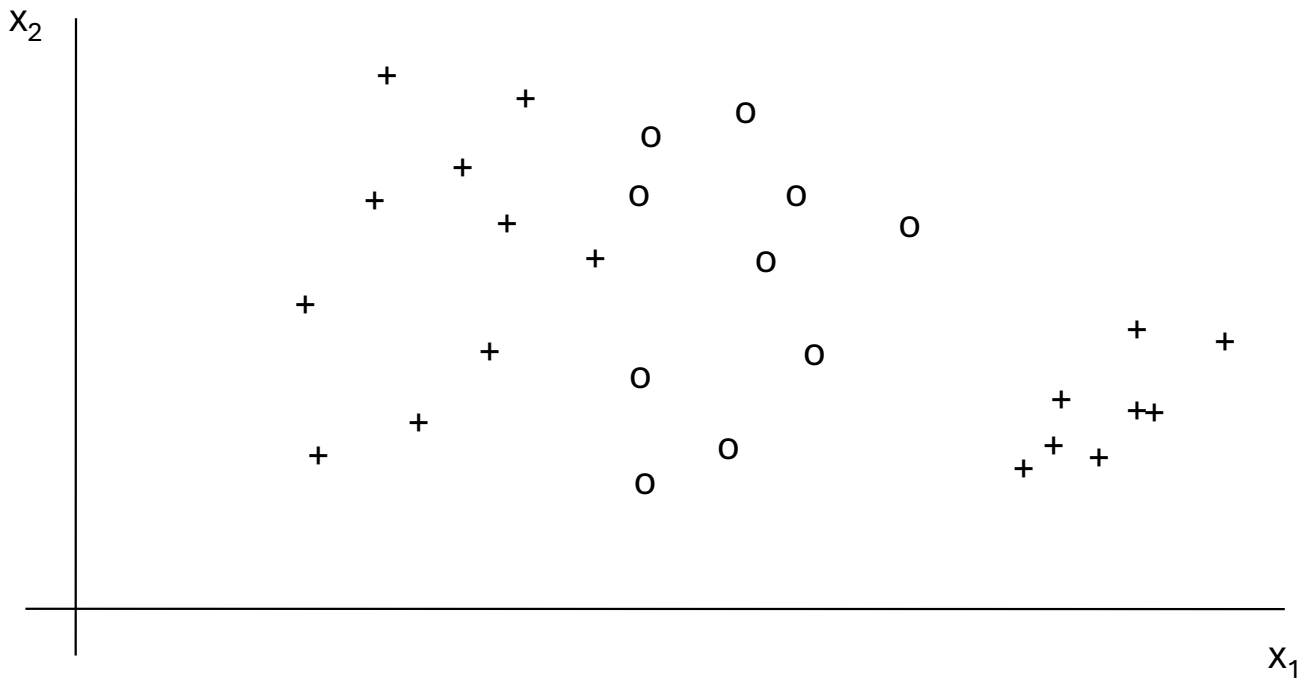
Instance-based learning

- Variants: *k*-NN with minimum distance:
  - With non-homogeneous sets the technique would fail:



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Instance-based learning

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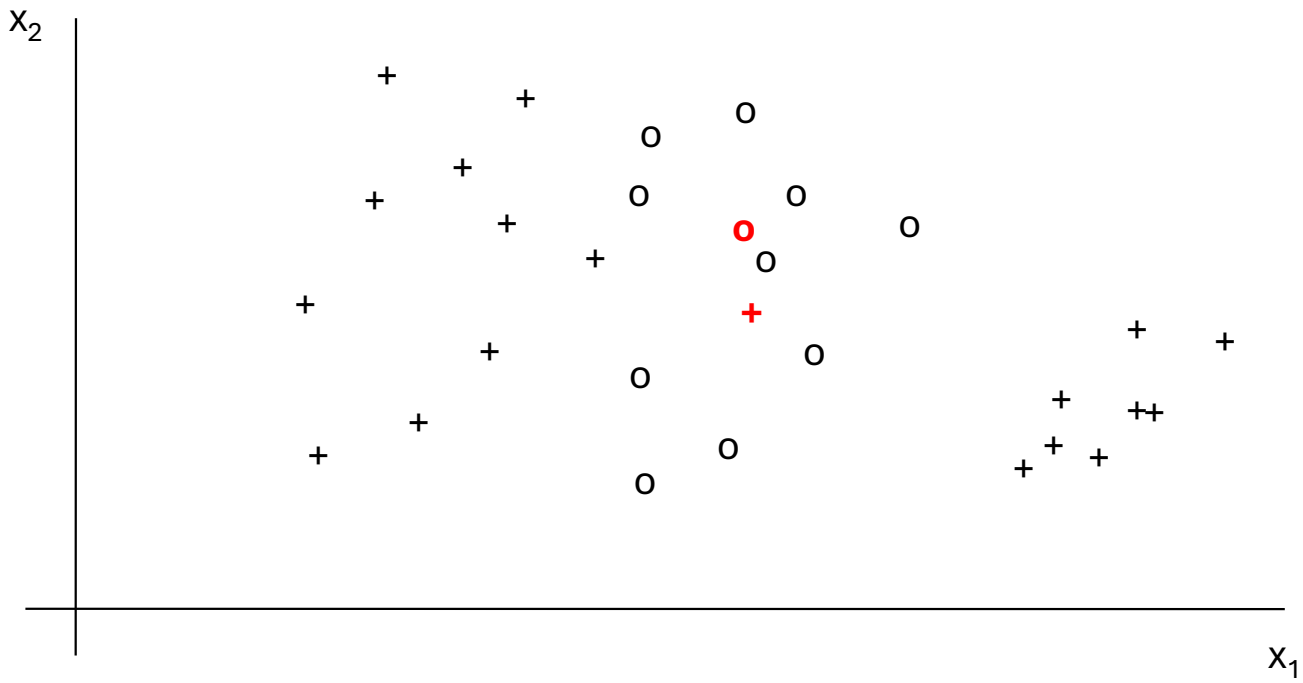

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Instance-based learning

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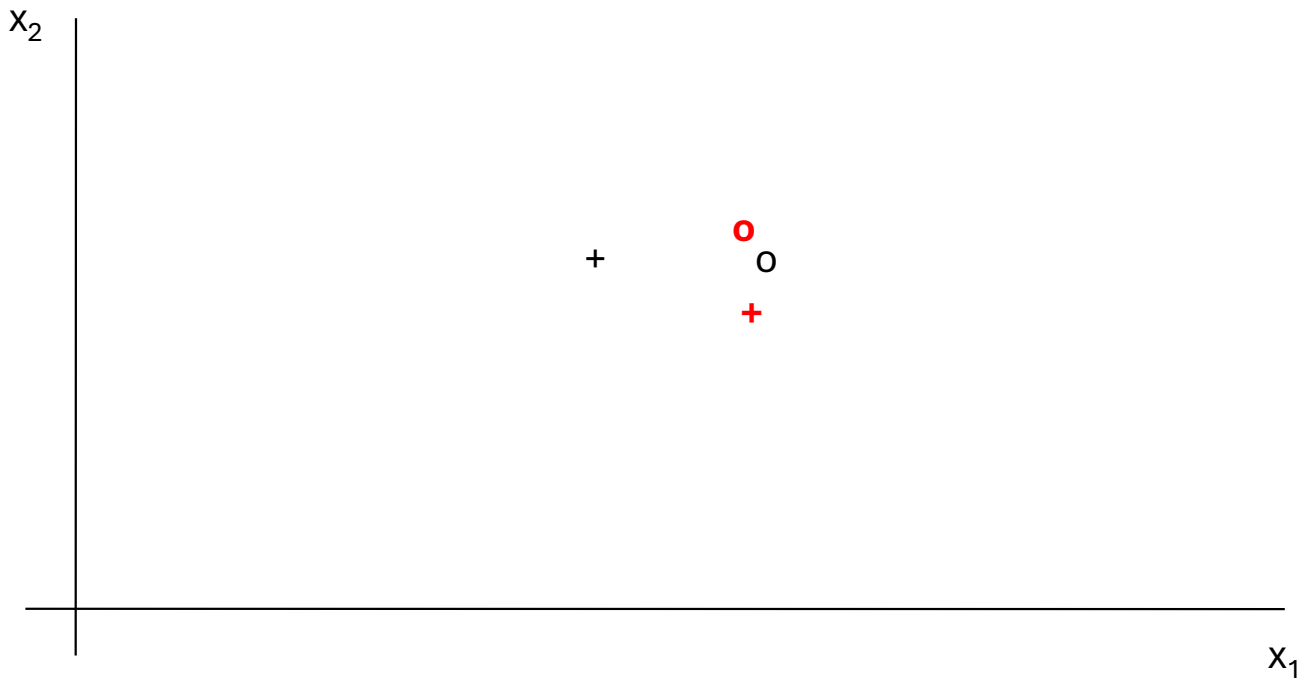

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Instance-based learning

- Variants: *k*-NN with minimum distance:
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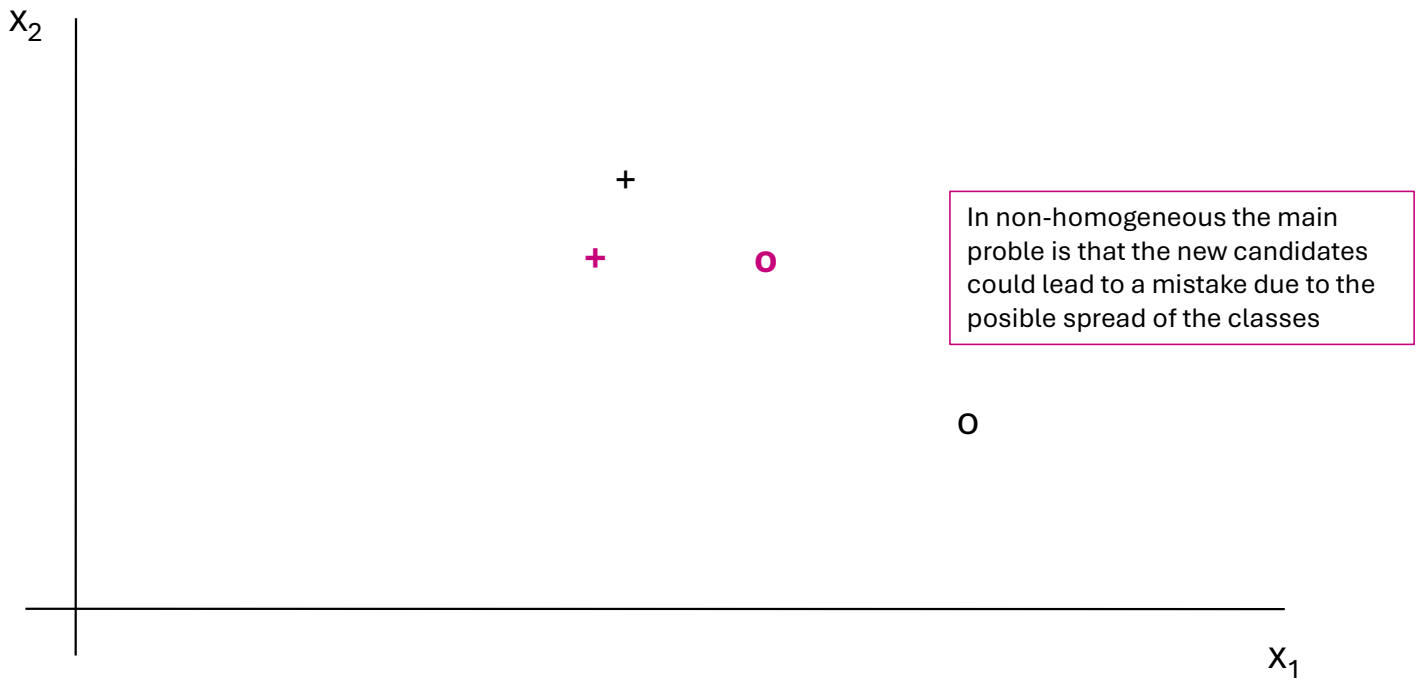

In non-homogeneous the main proble is that the new candidates could lead to a mistake due to the posible spread of the classes

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Instance-based learning

- Variants of k-NN:
  - *k*-NN with rejection
  - *k*-NN with average distance
  - *k*-NN with neighbour weighting
  - *k*-NN with minimum distance
  - ***k*-NN with weighting of variables**

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Instance-based learning

- Variants: *k*-NN with weighting of variables
  - Until now, the calculation of the distances weights all the variables equally
    - Are all the attributes or features equally relevant?
    - Does that relevance depend on the area of space?
  - If there are, for example, 20 variables, and only two of them are relevant, instances that are actually very different may be very close in space
    - "Curse of Dimensionality"

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Instance-based learning

- Variants: *k*-NN with weighting of variables
  - Until now, the calculation of the distances weights all the variables equally
    - Euclidean distance :

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^{n} (x_i[r] - x_j[r])^2}$$

    - n: number of variables

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Instance-based learning

- Variants: k-NN with weighting of variables
  - The Euclidean distance

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^{n} (x_i[r] - x_j[r])^2}$$

  - gives equal weight to all variables
    - It can be dangerous if there are any irrelevant variables
    - Solution: weight each of the variables
      - Corresponds to modifying the "length" of the axes in space
      - For example, with the above formula:

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^{n} w_r (x_i[r] - x_j[r])^2}$$

      - w_r assigns a weight to the variable r

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Instance-based learning

- Variants: k-NN with weighting of variables
  - Example:
    - Variable X_1 is irrelevant to C
    - Variable X_2 is relevant for C
  - How to calculate the w_r terms?
    - One way could be from mutual information measure $I(X_i, C)$ between the variable x_i and the target C-class variable

$$I(X_i, C) = \sum_{x_i, c} p_{(X_i, C)}(x_i, c) \log \frac{p_{(X_i, C)}(x_i, c)}{p_{X_i}(x_i) \cdot p_C(c)}$$

X_1	X_2	C
0	0	1
0	0	1
0	0	1
1	0	1
1	0	1
1	1	1
0	1	0
0	1	0
0	1	0
1	1	0
1	1	0
1	0	0

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Instance-based learning

- Variants: k-NN with weighting of variables
  - Measurement of mutual information between two variables
    - Reduction in uncertainty about one of the variables when the value of the other variable is known
    - The greater the measure of mutual information between the variables, the greater the "dependence" between them
    - In this case, the minimum information measure between each variable and the class variable is calculated
      - The weight wr associated with the variable xr will be proportional to the measure of mutual information $I(x_r, C)$

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Instance-based learning

- There are always problems when the dataset is very large:
  - Storage problems
  - Neighbourhood calculation problems

- Solutions:
  - Indexing
  - Selection of instances
  - Replacement of instances

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Instance-based learning

- There are always problems when the dataset is very large:
  - Storage problems
  - Neighbourhood calculation problems

- Solutions:
  - **Indexing**
  - Selection of instances
  - Replacement of instances

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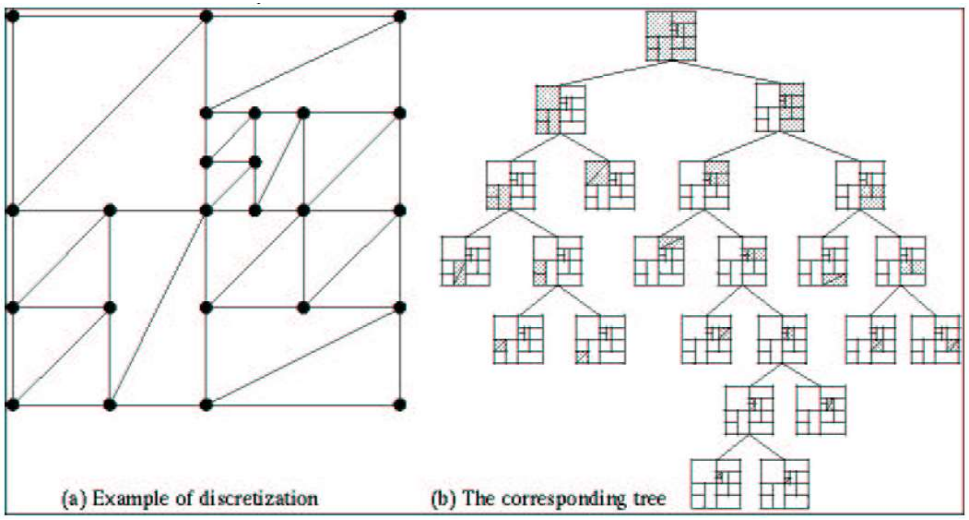
Instance-based learning

- Solutions: Indexing:
  - KD trees
    - A. W. Moore



  - Clustering
    - Unsupervised learning

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Instance-based learning

- There are always problems when the dataset is very large:
  - Storage problems
  - Neighbourhood calculation problems

- Solutions:
  - Indexing
  - **Selection of instances**
  - Replacement of instances

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Instance-based learning

- Solutions: Selection of instances:
  - Choose a small group of instances (prototypes) (S) that maintain the same information as the total set (T)
  - Methods:
    - Incremental:
      - Start with an empty prototype set S
      - Add instances to the set S from the instances in T, as long as they meet a certain criterion
        - For example, when trying to classify it, it is classified differently from its class
          - Hart Condensation

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Instance-based learning

- Solutions: Selection of instances:
  - Choose a small group of instances (prototypes) (S) that maintain the same information as the total set (T)
  - Methods:
    - Decremental:
      - Start with a set $S = T$ of prototypes
      - Remove instances from the set S, as long as they meet a certain criteria
        - For example, that when extracted from the set, it is still classified correctly
          - Wilson Edition

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Instance-based learning

- There are always problems when the set of instances is very large:
  - Storage problems
  - Neighbor calculation problems
- Solutions:
  - Indexing
  - Selection of instances
  - **Replacement of instances**

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Instance-based learning

- Solutions: Replacement of instances:
  - Calculate **prototypes** from the training set
  - Normally 1-NN
  - Learning Vector Quantization (LVQ):
    - Start with a set of prototypes $S = \{s_1, \dots, s_M\}$
    - Repeat:
      - Choose a new instance x
      - Get the closest prototype of S, $s = argmin_i\ (d(x, s_i))$
      - Update the position of s_i:

$$s_i = s_i + \alpha\,[x - s_i] \text{ if } s_i \text{ and } x \text{ belong to the same class}$$
$$s_i = s_i - \alpha\,[x - s_i] \text{ if } s_i \text{ and } x \text{ belong to different classes}$$

  - LVQ problem: Definition of the number of prototypes to use

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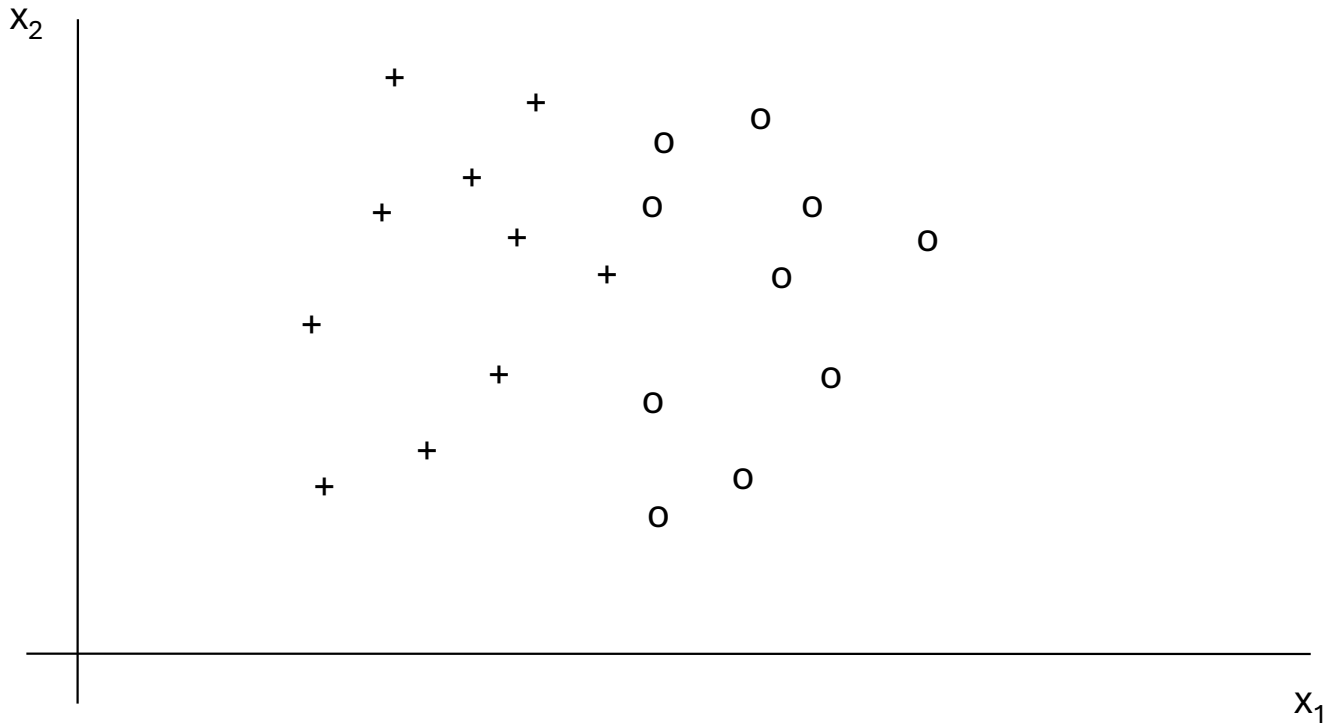
Instance-based learning

- Solutions: Replacement of instances:
  - Dataset:



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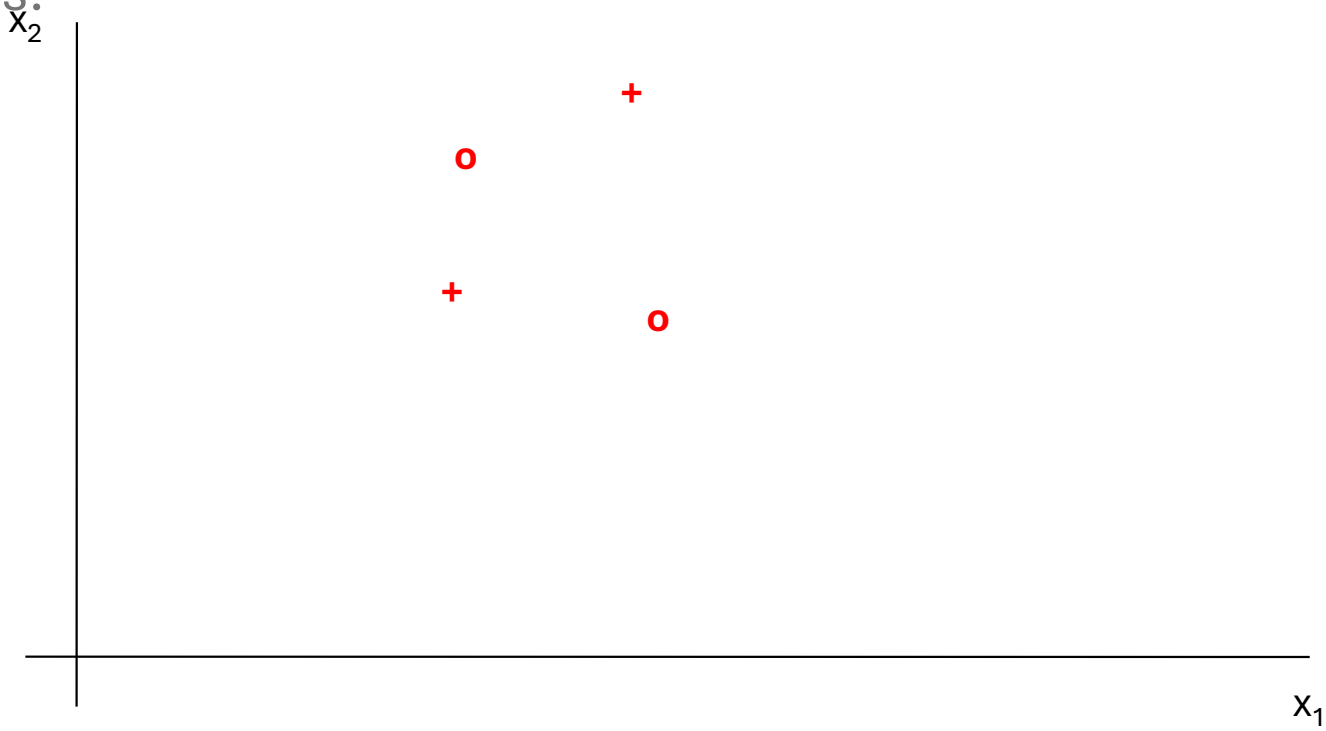
Instance-based learning

- Solutions: Replacement of instances:
  - Prototypes:



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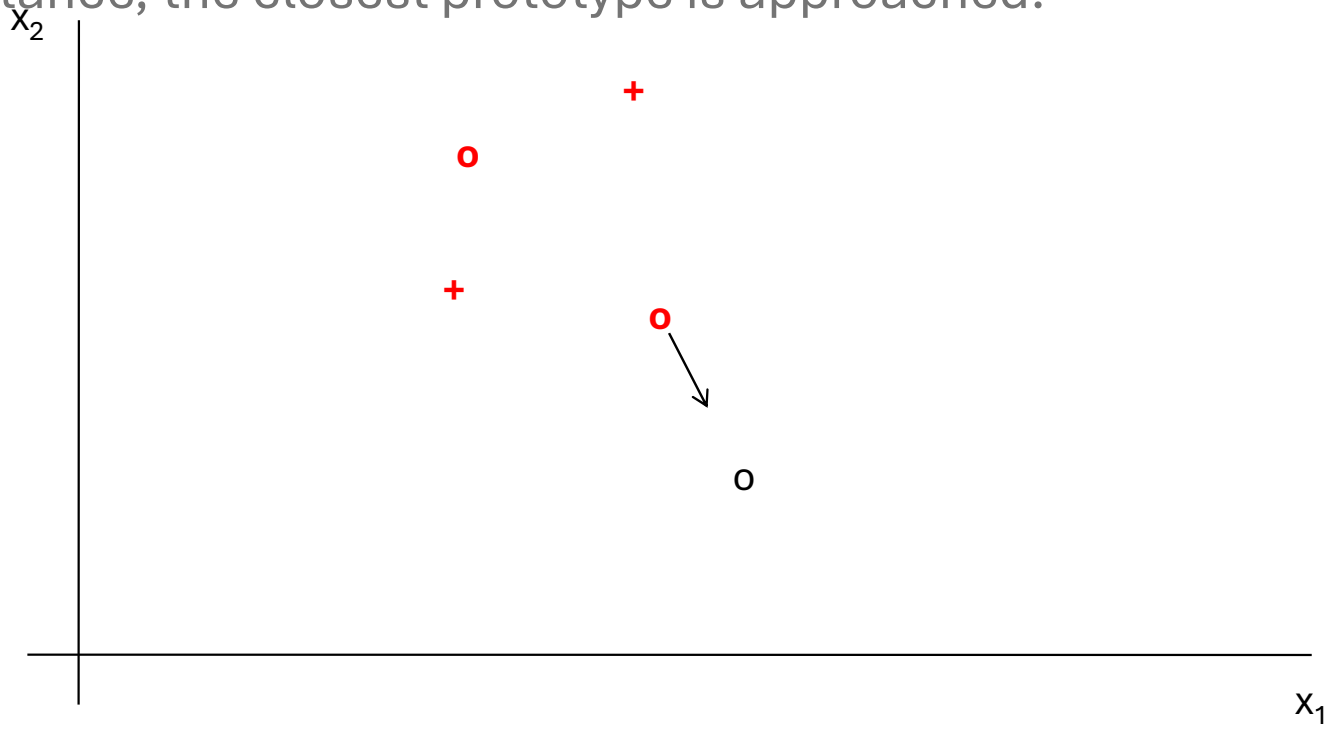
Instance-based learning

- Solutions: Replacement of instances:
  - In this instance, the closest prototype is approached:



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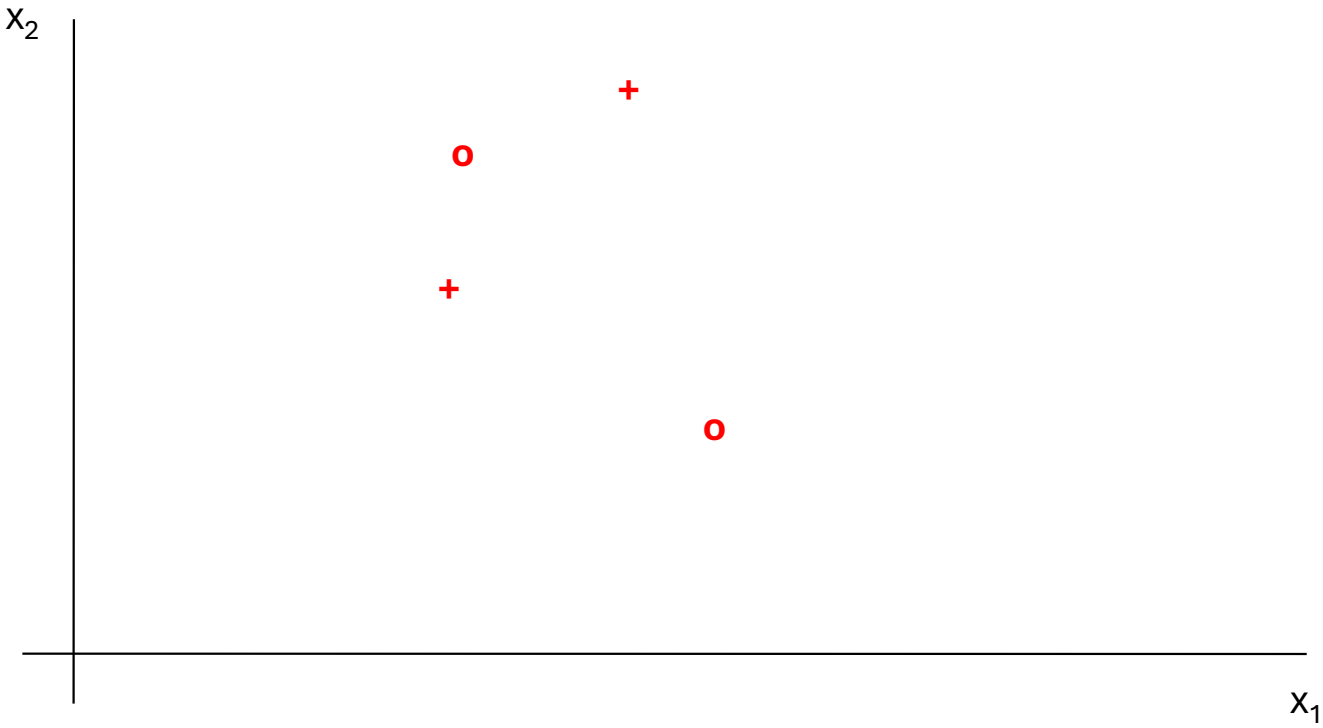
Instance-based learning

- Solutions: Replacement of instances:



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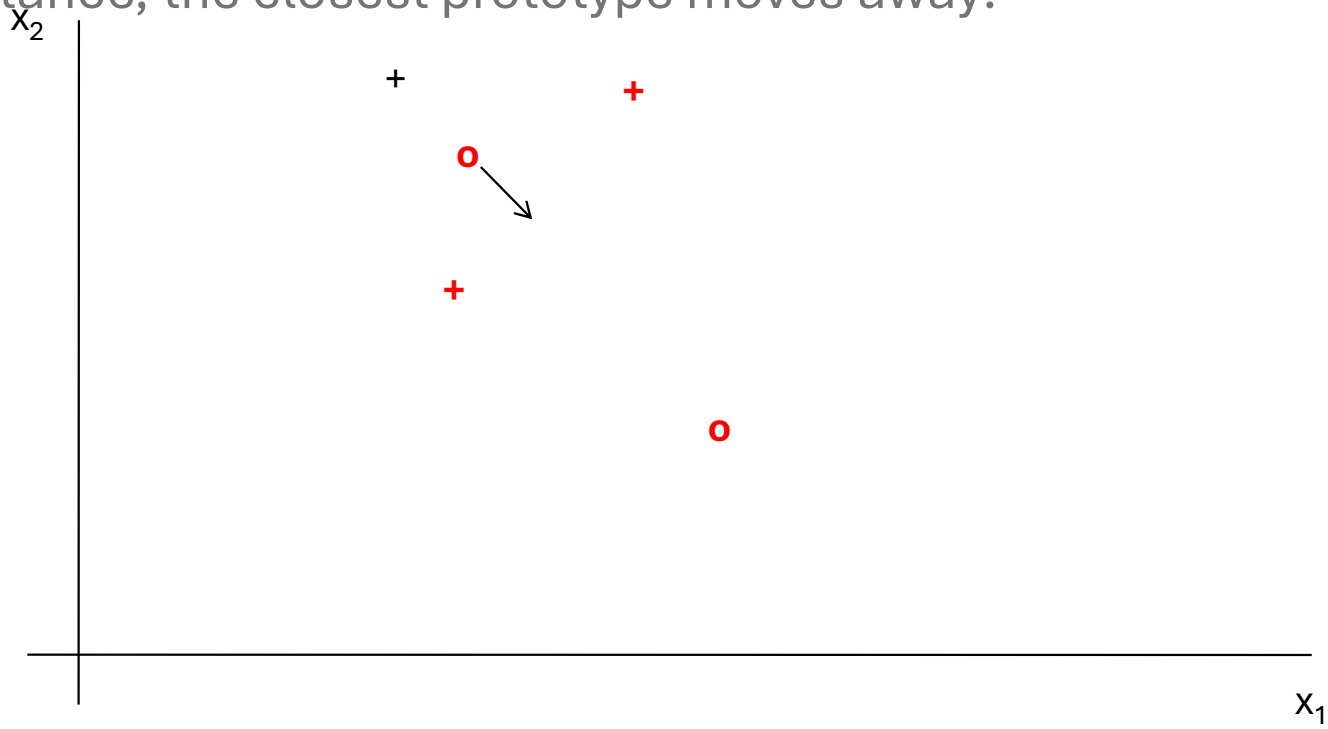
Instance-based learning

- Solutions: Replacement of instances:
  - In this instance, the closest prototype moves away:



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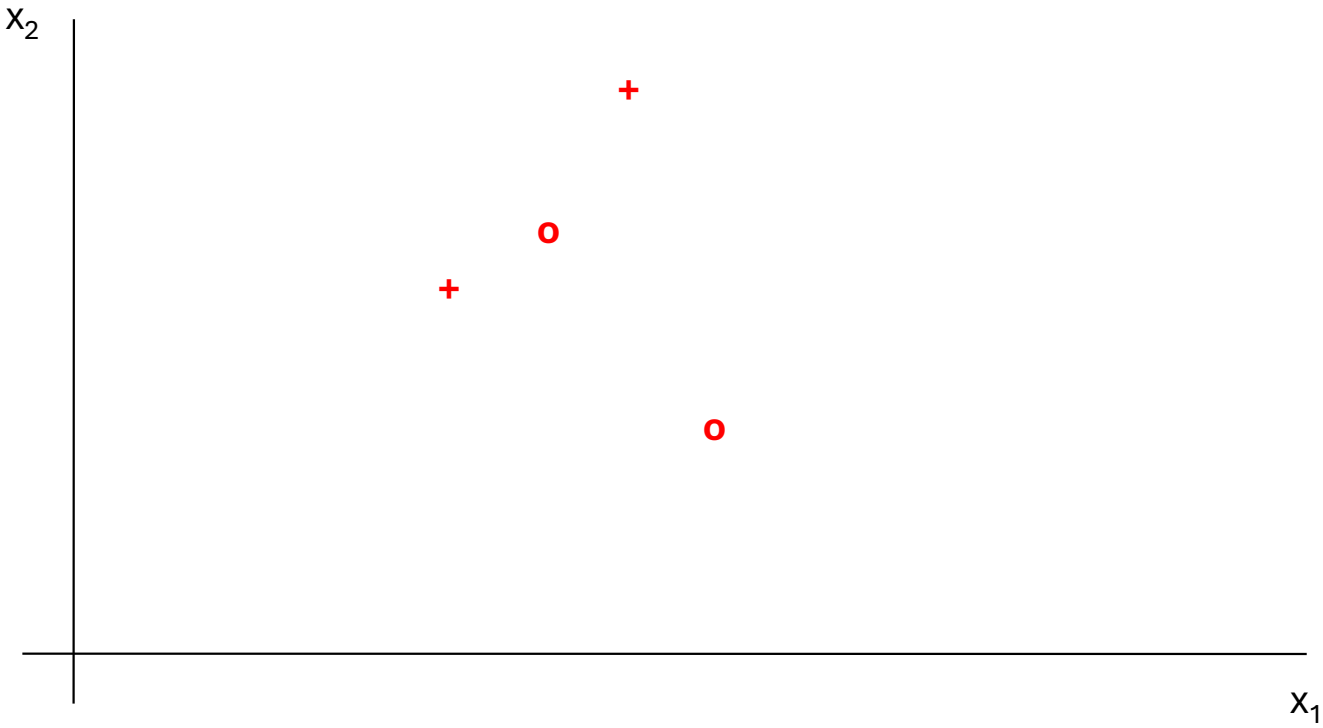
Instance-based learning

- Solutions: Replacement of instances:



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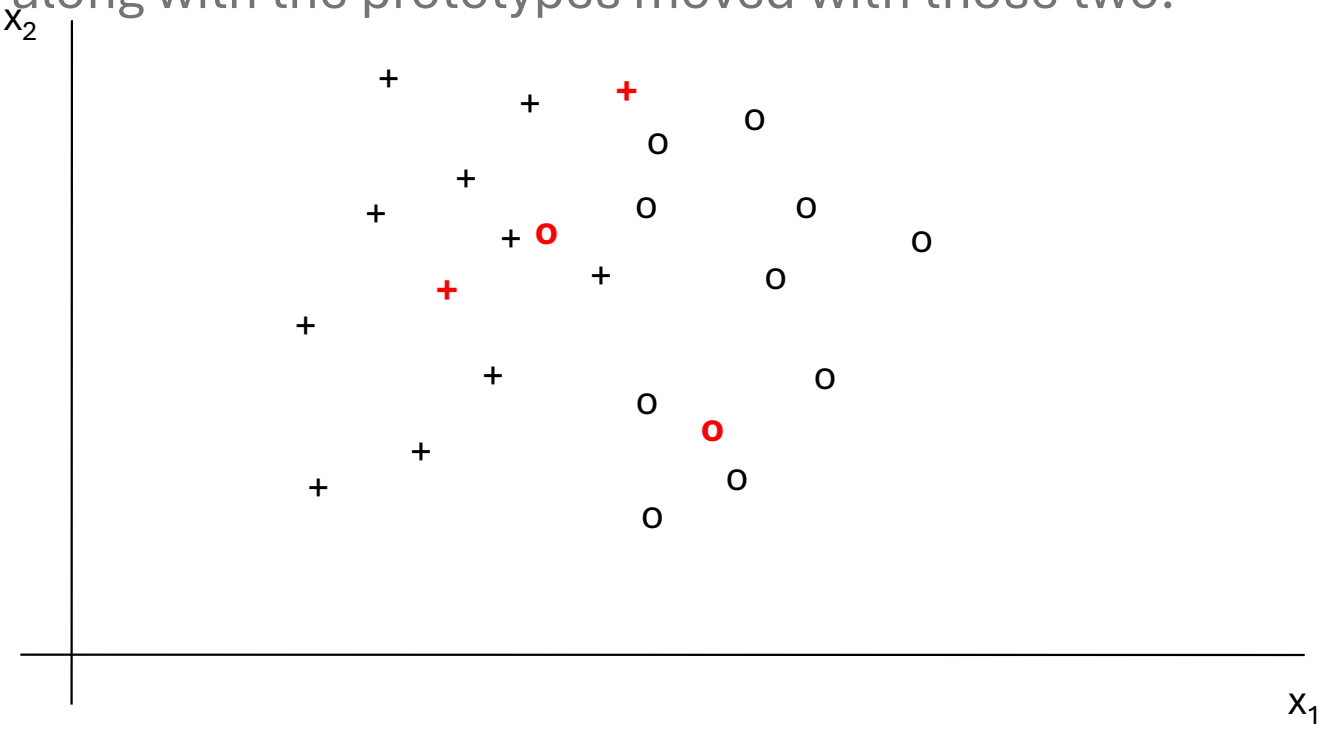
Instance-based learning

- Solutions: Replacement of instances:
  - Instances along with the prototypes moved with those two:



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Instance-based learning

- Related Methods:
  - Hybrid local models
  - Case-Based Reasoning

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Instance-based learning

- Related Methods:
  - **Hybrid local models**
  - Case-Based Reasoning

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Instance-based learning

- Related Methods:
  - Hybrid local models
    - If the space is complex, the desired response may not "behave" in the same way throughout the space.
      - Why make a single model?
    - Building local models
      - Locality not in areas of space, but in points
      - Created when querying a new case
    - When a new query is made, the k neighbours are obtained and the model is trained with them
      - Afterwards, the model is applied and deleted
        - It is only valid at that point

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Instance-based learning

- Related Methods:
  - Hybrid local models
    - Regression trees and regression model trees
      - Local models applying to an area
    - kNN + model
      - Local model applying only to a single point in space

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Instance-based learning

- Related Methods:
  - Hybrid local models
    - Regression:
      - Instead of returning the average value of the targets of the nearest neighbours, develop a linear regression model:

$$\hat{f}(x) = a_0 + a_1 x[1] + \ldots + a_n x[n]$$

      - Once you have it, apply it, return the output and remove the model
        - Given a new instance, build another model.

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Instance-based learning

- Related Methods:
  - Hybrid local models
    - Regression:
      - If a linear model is created, it is assumed that each point can be approximated locally by a linear function
      - Applicable with any other approximation method:
        - Non-linear model: quadratic, cubic, etc.
        - Neural networks
        - Etc
      - Low complexity models
        - Why?

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Instance-based learning

- Related Methods:
  - Hybrid local models
    - Classification:
      - Instead of a vote on nearest neighbour classes, a low complexity classifier is trained
        - Why low complexity?
      - For example, a linear SVM
        - SVM-KNN classifier
        - Hypothesis: locally, data is linearly separable

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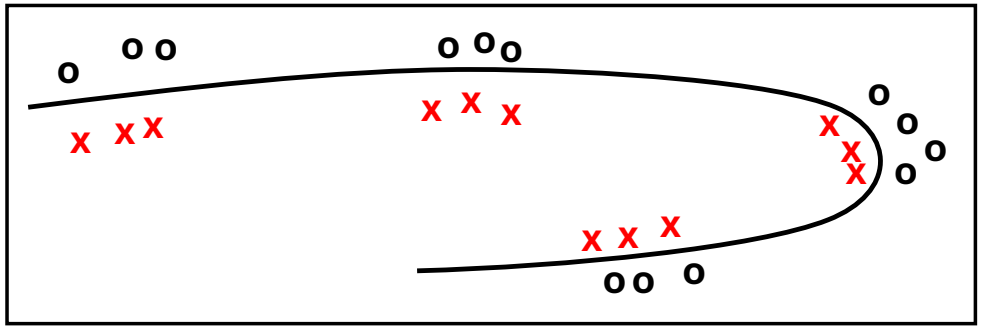
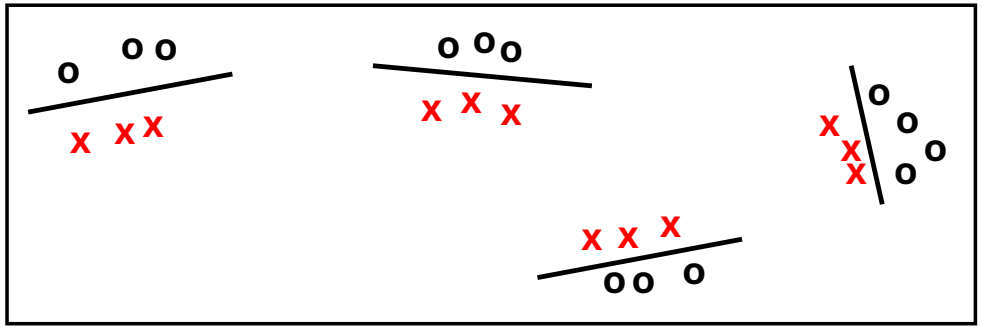
Instance-based learning

- Related Methods:
  - Hybrid local models
    - Classification:
      - Global model:



      - Local models:



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Instance-based learning

- Related Methods:
  - Hybrid local models
    - Advantages:
      - When faced with a complex space, it can be complex to make a global model
        - Local models can be a good approximation
    - Disadvantages:
      - More computational needs
        - Calculation of nearest neighbours
        - Model development

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Instance-based learning

- Related Methods:
  - Hybrid local models
  - **Case-Based Reasoning**

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - What happens when the instances are represented in a more complex way?
    - Example:
      - Transport:<bus>
      - Weather:<rainy>
      - Forecast:<day+1: cloudy, day+2=sunny, day+3=?>
      - Location:<house[rooms=2, no pool], center=far>
      - People:<adults[men=1, women=1], children=4>

      - Satisfaction: ????

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - CBR (Case-Based Reasoning)
    - It is an Artificial Intelligence technique that is based on the use of previous experiences to solve new problems by means of the following hypothesis:
      - Similar problems have similar solutions
      - Typical in human learning
        - Learning by analogy

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - Given a problem to be solved, the CBR searches, in a database called the Case Base, for similar problems that have previously been successfully solved, called cases, and adapts the solutions to provide a solution to the current problem
    - This reasoning mechanism is used by humans in multiple problems and makes it an easy to understand system

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - The CBR involves a whole methodology with a loop of activities that, in addition to solving new problems, allows us to learn from the good solutions obtained for the new problems:
      - Retrieve
      - Reuse
      - Check
      - Retain

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - Retrieve:
      - Given a problem, the most similar cases are retrieved from the Case Base
        - A case is a previous problem with its solution
    - Reuse:
      - Extract the solution from the selected case for use
      - This may involve adapting the solution to the new situation.

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - Check:
      - It must be analyzed if the new solution is acceptable and if it is necessary to revise it
      - Although it is usually understood that the reuse process involves the whole problem of adapting the recovered case(s) to the new problem, in many practical applications the reuse and revise phases are hardly distinguished
        - Many researchers speak of **adaptation** phase, which combines both
    - Retain:
      - After the solution has been successfully applied, the experience must be stored as a new case in the Case Base

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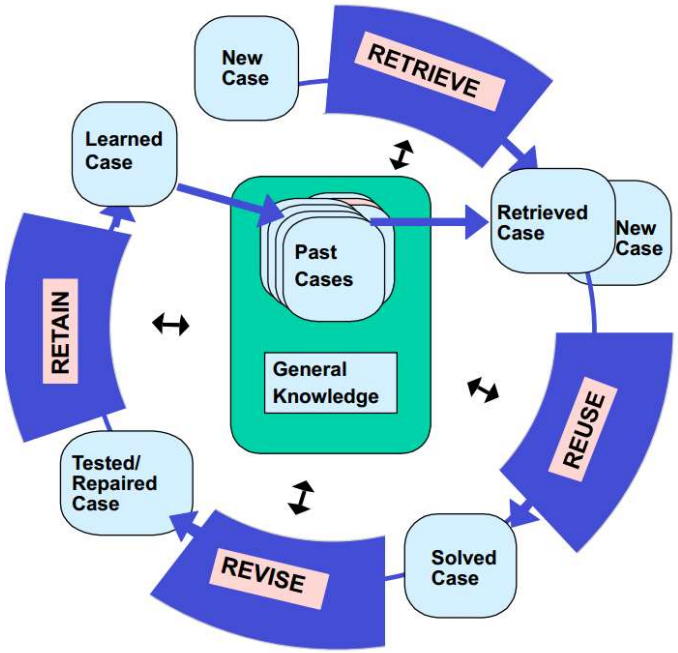
Instance-based learning

- Related Methods:
  - Case-Based Reasoning



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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - The four processes are not unique tasks, that is, each of them involves carrying out a series of more specific tasks.
      - Task hierarchy

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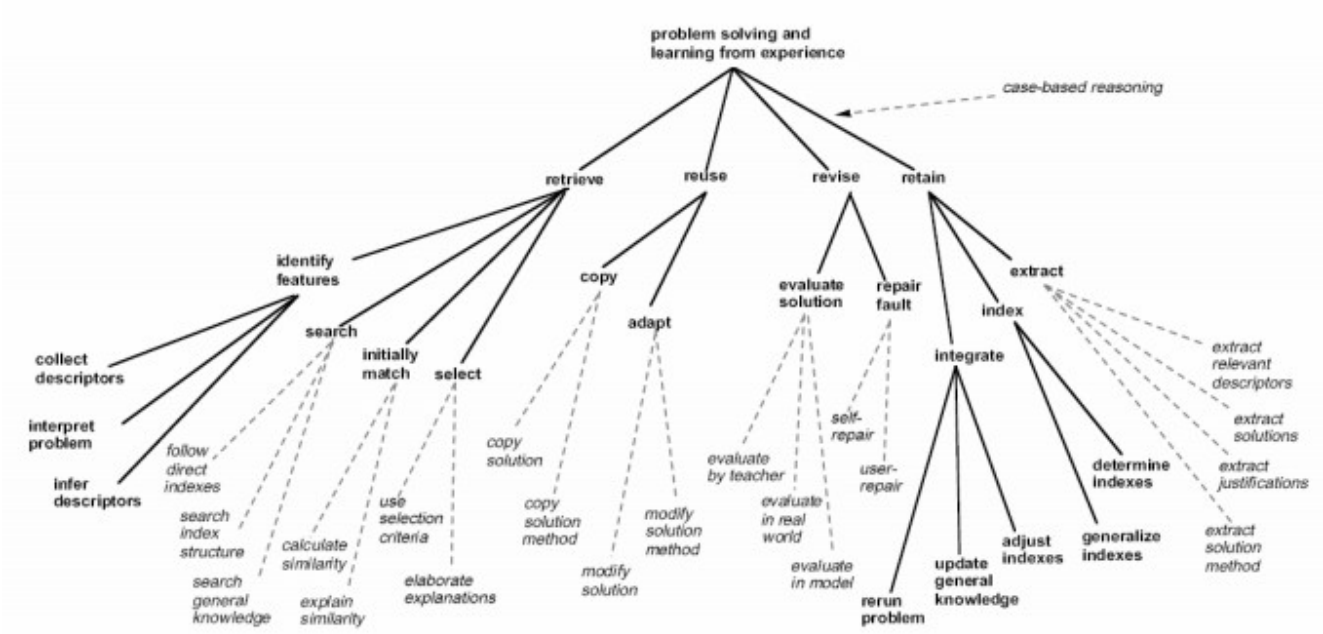
Instance-based learning

- Related Methods:
  - Case-Based Reasoning



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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - Based on the same principles as kNN
      - An instance is classified based on similar cases
      - The difference is that instead of using points in an Euclidean space, we represent the instances with more complex attributes
    - Look for a similarity metric that depends on the working domain
    - The solution is based on complex, application-domain specific combinations

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Instance-based learning

- Related Methods:
  - Case-Based Reasoning
    - Much more complex representations of information:
      - Images
      - documents
      - plans
      - etc.

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Instance-based learning

- Paradigm that can be used in other types of spaces:
  - Instead of instances represented in an Euclidean space
    - More complex representations
  - For example: image classification:
    - Dataset with images of different classes
    - By creating a model (ANN, SVM, etc):
      - Extract features from images
      - Create dataset with those features and classes
      - Fit a machine learning model
      - Given a new instance, extract features and apply the model

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Instance-based learning

- Paradigm that can be used in other types of spaces:
  - Instead of instances represented in an Euclidean space
    - More complex representations
  - For example: image classification:
    - Dataset with images of different classes
    - Using the instance-based learning paradigm:
      - Develop a function that, between two images, gives a measure of similarity between them
      - Given a new image, calculate the closest ones and from them, calculate the membership class

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Instance-based learning

- Lazy algorithms vs. greedy algorithms
  - Lazy algorithms [KNN, Local Regression…] delay the calculation of a hypothesis until the arrival of a new query
    - They compute a local approximation of the objective function to answer each new query
    - In other words, they use multiple local approximations to model the [global] objective function
  - Greedy algorithms can also use approximations; however, they remain “fixed” to the training set.
    - A model is made
  - Given the same hypotheses space, lazy algorithms have a greater power of adaptation to a new query
    - and greater adaptation to changes in the problem

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Instance-based learning

- Summary:
  - These methods delay the creation of a hypothesis until the moment of a new classification
  - This allows them to generate a **local approximation** for each of the new instances, approximating the target with complex functions
  - Disadvantages:
    - The cost of calculating these approximations
    - Defining an appropriate metric
    - Storage

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